Knightian Uncertainty and Fair Value Estimation

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Abstract

We examine whether managers' fair value estimates are affected by Knightian uncertainty, which arises when the probability distribution governing future outcomes is unknown or not uniquely specified. Theory predicts two countervailing effects of Knightian uncertainty on fair value estimates. First, managers wait and see, delaying downward revisions to reported asset values. Second, they place greater weight on probability models that imply adverse outcomes, producing more pessimistic fair value estimates. Consistent with these predictions, managers are less likely to recognize goodwill impairments when Knightian uncertainty is greater and report larger impairments conditional on recognition. Our findings also suggest that mutual fund managers report lower corporate bond valuations when Knightian uncertainty is greater. Finally, we find evidence that suggests investors respond less to a given goodwill impairment amount when Knightian uncertainty is greater and that funds receive lower subsequent inflows when markdowns are concentrated in higher-uncertainty holdings. Our findings suggest that Knightian uncertainty systematically alters the timing and level of reported asset values and affects how investors respond to them.

1. Introduction

Fair value estimation requires managers to exercise discretion when forecasting cash flows, selecting discount rates, and determining other valuation inputs. Prior research provides evidence that some managers use this discretion opportunistically to bias their fair value estimates (Hanley et al., 2018). Other research suggests that behavioral biases, such as cost anchoring, affect how managers exercise this discretion and, in turn, the fair value estimates they produce (Anderson et al., 2025). We examine whether Knightian uncertainty affects managers' fair value estimates.

Economists distinguish between two forms of uncertainty: risk and Knightian uncertainty. Risk is uncertainty about the eventual outcome when the probability distribution over possible outcomes is known. Knightian uncertainty is uncertainty about the probability distribution itself, which is unknown or not uniquely specified (Knight, 1921). This distinction matters because agents respond differently to Knightian uncertainty than to risk. Agents averse to Knightian uncertainty consider multiple probability models consistent with the available information and place greater weight on models that imply adverse outcomes (Ellsberg, 1961; Gilboa and Schmeidler, 1989; Halevy, 2007). We build on this theory and predict that this asymmetric belief formation leads managers to report lower fair value estimates. This possibility is important because managers' aversion to Knightian uncertainty may result in reported fair values that are more pessimistic than the underlying economic value of the asset or liability would warrant.

The FASB's conceptual framework identifies neutrality as a component of faithful representation and describes neutral information as information that is not slanted, weighted, emphasized, or otherwise presented to influence how users receive it (FASB, 2018). Knightian uncertainty can challenge neutrality if managers systematically give greater weight to adverse scenarios, causing reported values to disproportionately reflect pessimistic states. This concern is particularly salient when fair value estimates require managers to exercise greater discretion. ASC 820 classifies fair value inputs into a three-level hierarchy based on their observability (FASB, 2006). Level 1 inputs are quoted prices in active markets and leave little room for managerial judgment. Level 2 inputs are observable only indirectly and require managers to select and weight comparable market information. Level 3 inputs are unobservable and require model-based assumptions and estimates. Thus, for Level 2 and Level 3 estimates, managers must select and weight valuation inputs, assumptions, and scenarios. Under Knightian uncertainty, disproportionate weight on adverse scenarios can result in more pessimistic reported fair values.

In this paper, we investigate whether and how Knightian uncertainty affects managers' fair value estimates. We focus on Level 2 and Level 3 fair value estimates. For the Level 3 setting, we examine goodwill impairment testing, which requires managers to project cash flows under alternative business conditions, select discount rates, estimate the reporting unit's fair value, and determine whether and by how much goodwill is impaired. For the Level 2 setting, we study mutual funds' valuations of corporate bonds. Because many corporate bonds held by mutual funds do not trade actively, they are valued using observable but indirect inputs, such as yield curves, credit spreads, and evaluated prices. These estimates require substantial judgment in selecting comparable securities and adjusting for uncertainty.

A central empirical challenge is separating Knightian uncertainty from risk and from widely used proxies for general uncertainty (e.g., the VIX, policy uncertainty, macro-uncertainty, and disagreement). These proxies are useful for many questions, but they do not isolate the Knightian uncertainty faced by a representative agent, as described by Knight (1921), Ellsberg (1961), and Halevy (2007). We follow recent developments in the expected utility with uncertain probabilities (EUUP) literature, which links Knightian uncertainty to uncertainty about probabilities rather than the volatility of outcomes (Izhakian, 2017, 2020). Specifically, we use firm-level Knightian uncertainty estimates constructed from intraday TAQ returns following Izhakian (2017, 2020). This measure is outcome-independent, risk-independent, and preference-independent.

Our first test examines the relation between Knightian uncertainty and managers' goodwill impairment decisions. GAAP requires managers to make two distinct decisions: (1) whether to recognize an impairment and (2) conditional on recognition, the amount to recognize (Glaum et al., 2018). Because write-downs are quasi-irreversible, real options theory predicts that managers facing greater Knightian uncertainty will delay impairment recognition to preserve the option to wait (Roychowdhury and Martin, 2013). Conditional on recognition, however, theory further predicts that Knightian-uncertainty-averse managers will place greater weight on adverse valuation outcomes and therefore record larger impairments (Epstein and Schneider, 2003; Hansen and Sargent, 2008). Together, these forces imply that Knightian uncertainty lowers the likelihood of recognizing an impairment (the extensive margin) but raises its magnitude once recognized (the intensive margin).

We begin by estimating the relation between Knightian uncertainty and goodwill impairments, controlling for expected risk and other firm characteristics. We then model the impairment decision using a two-stage hurdle model (Cragg, 1971; Nessa, 2017). The first stage estimates whether Knightian uncertainty is associated with the likelihood of recognizing a goodwill impairment. Conditional on recognizing an impairment, the second stage estimates whether Knightian uncertainty is associated with the magnitude of the impairment. We find that managers are less likely to recognize a goodwill impairment when Knightian uncertainty is greater. This evidence is consistent with managers delaying impairment recognition until uncertainty resolves. However, conditional on recognition, greater Knightian uncertainty is associated with larger impairments. This finding is consistent with managers incorporating more pessimistic forecasts into the fair value estimates used to measure goodwill impairments. Taken together, the results suggest that Knightian uncertainty reduces the likelihood of recognizing an impairment but increases the magnitude of recognized impairments. The positive unconditional association is consistent with larger conditional impairments outweighing the lower likelihood of recognition.

To mitigate the concern that the documented relation between Knightian uncertainty and unconditional goodwill impairment charges is driven by firm fundamentals rather than managers' aversion to Knightian uncertainty, we examine whether the relation between Knightian uncertainty and managers' fair value estimates is stronger when managers are given greater discretion. We use ASU 2011-08 as the setting because it granted managers greater discretion in goodwill impairment testing. The standard introduced an optional qualitative screen, commonly referred to as "Step 0," and eliminated the limited ability to carry forward a prior-period fair value calculation (FASB, 2011). Managers who elect Step 0 assess and weigh adverse and mitigating factors to determine whether a quantitative impairment test is necessary. If Knightian uncertainty leads managers to place greater weight on adverse qualitative factors, the additional discretion provided by the standard should strengthen the relation between Knightian uncertainty and fair value estimates. Because ASU 2011-08 operates through the qualitative assessment that determines whether the quantitative impairment test is performed, its most direct effect is on the recognition decision. We therefore examine whether the relation between Knightian uncertainty and impairment recognition changed after adoption, and whether any change was stronger among firms with greater predetermined goodwill exposure.[1] We find that, following adoption, greater Knightian uncertainty is associated with a lower likelihood of impairment recognition among firms with greater predetermined goodwill exposure. Because adoption is common to all firms rather than staggered across treated and untreated units, we interpret this pattern as associative evidence consistent with Knightian uncertainty affecting recognition through managers' judgment rather than as causal evidence, and it helps mitigate the concern that the documented relation merely reflects firm fundamentals.

Our second set of tests focuses on Level 2 fair value estimation by studying mutual fund managers' valuations of corporate bond holdings reported quarterly on Form N-PORT (SEC, 2016). Corporate bonds are periodically valued by mutual fund managers using observable

[1]We focus on recognition because ASU 2011-08 operates through the qualitative assessment that governs whether the quantitative impairment test is performed. The analogous test for the conditional magnitude of impairments shows no change, so we do not interpret the standard as affecting the amount recognized conditional on recognition.

but indirect inputs, including yield curves, credit spreads, evaluated pricing, and matrix pricing, leaving substantial scope for judgment in selecting comparables and adjusting inputs to reported values (Cici et al., 2011). We focus on infrequently traded corporate bonds and examine the extent to which fund managers mark down their valuations of these bonds relative to benchmark prices. Using a within-fund-time design, we find that mutual fund managers report lower valuations relative to market-based benchmark prices for bonds issued by firms with greater Knightian uncertainty. These results suggest that Knightian uncertainty affects managers' Level 2 fair value estimates, resulting in more pessimistic reported values.

Finally, we investigate the economic consequences of fair value estimates formed under Knightian uncertainty. In the Level 3 setting, we examine whether investors react less to goodwill impairment amounts when Knightian uncertainty is greater. We find a weaker market response to a given impairment amount under greater Knightian uncertainty, consistent with investors placing less weight on impairment estimates that may reflect both underlying fundamentals and pessimistic valuation judgments. In the Level 2 setting, we test whether subsequent fund flows vary when reported bond markdowns are concentrated in holdings with greater Knightian uncertainty. We find that such funds receive lower subsequent net flows and inflows, consistent with investors responding less favorably to markdowns concentrated in holdings with greater Knightian uncertainty. Overall, our evidence suggests that investors place less weight on, and respond less favorably to, fair value estimates formed when there is greater Knightian uncertainty.

The effects of Knightian uncertainty on impairment recognition and measurement are conceptually distinct from conditional conservatism. Conditional conservatism is typically framed as an asymmetric recognition rule that produces timelier recognition of losses than gains when bad news is more verifiable or litigation and contracting incentives are stronger. Knightian uncertainty, in contrast, generates two distinct predictions. The option value of waiting leads managers to delay quasi-irreversible write-downs, reducing the likelihood of impairment recognition. Conditional on recognition, aversion to Knightian uncertainty leads managers to place greater weight on adverse states, resulting in more pessimistic valuations and larger impairments. Our findings are consistent with this pattern. Greater Knightian uncertainty is associated with less frequent impairment recognition but larger impairments conditional on recognition. Our corporate bond tests provide complementary evidence from a Level 2 setting in which no discrete loss-recognition decision analogous to conditional conservatism exists. Taken together, the findings support the distinct recognition and measurement predictions of Knightian uncertainty.

We contribute to the extensive literature on fair value estimation and managerial discretion. Prior research primarily examines fair value discretion through the lens of agency theory, asking whether managers use judgment to convey private information or to opportunistically manage earnings (e.g., Ramanna and Watts, 2012; Hanley et al., 2018; Chung and Hribar, 2021). We introduce managers' aversion to Knightian uncertainty as a distinct influence on fair value estimates that operates alongside these incentives. By documenting that greater Knightian uncertainty is associated with more pessimistic fair value estimates, we identify a countervailing force that may naturally constrain opportunistic inflation.

Second, we extend the emerging literature on the economic consequences of Knightian uncertainty. While recent studies in finance have shown that Knightian uncertainty influences real corporate decisions such as investment timing and capital structure (e.g., Izhakian and Yermack, 2017; Izhakian et al., 2022), limited evidence exists regarding its relation to financial reporting activities. We bridge this gap by providing evidence that suggests Knightian uncertainty affects managers' fair value estimates in both Level 2 and Level 3 settings.

Third, we provide evidence on the interplay between real options theory and accounting recognition. Prior research links uncertainty-driven delays in recognition to discretionary conservatism (e.g., Roychowdhury and Martin, 2013). By separating the goodwill impairment decision into recognition and measurement stages, we distinguish between two theoretical mechanisms: the "option to wait," which delays recognition, and aversion to Knightian uncertainty, which produces more pessimistic estimates conditional on recognition. This distinction clarifies why greater Knightian uncertainty can reduce the likelihood of recognition while increasing the magnitude of recognized impairments.

Finally, our findings are informative to standard setters and regulators. The FASB's conceptual framework emphasizes neutrality as a core attribute of faithful representation. Our results suggest that neutrality may be more difficult to achieve when fair value estimates are formed under greater Knightian uncertainty. Neutrality may therefore depend in part on the information environment and verification infrastructure surrounding fair value estimates. Greater transparency around valuation inputs and assumptions, stronger external benchmarking, and enhanced verifiability may help users evaluate how managers arrive at reported values.

2. Theory and hypothesis development

2.1. Knightian uncertainty and aversion to Knightian uncertainty

Standard economic theory distinguishes between risk, where the probability distribution of future outcomes is known, and Knightian uncertainty (or ambiguity), where the distribution itself is unknown or non-unique (Knight, 1921; Ellsberg, 1961). In a risk setting, rational agents maximize expected utility based on a single, additive probability measure that fully characterizes the likelihood of all states. In contrast, agents facing Knightian uncertainty must process two distinct layers of uncertainty: the aleatory uncertainty regarding the stochastic nature of outcomes, and the epistemic uncertainty regarding the correct probability distribution assigned to those outcomes (Klibanoff et al., 2005).

The existence of Knightian uncertainty suggests that agents cannot identify a single objective prior. Instead, they entertain a set of plausible probability measures consistent with the available information (Gilboa and Schmeidler, 1989). A large literature documents that individuals exhibit a distinct aversion to Knightian uncertainty, preferring settings in which probabilities are objectively known to those in which they are vague (Epstein and Schneider, 2010). Under Knightian uncertainty aversion, agents place relatively more weight on probability density functions that predict unfavorable outcomes and less weight on those that predict favorable outcomes (Gilboa and Schmeidler, 1989). Effectively, the agent acts as if the likelihood of adverse states is higher than a neutral Bayesian update would suggest (Epstein and Schneider, 2003).

The literature suggests that corporate managers exhibit a significant aversion to Knightian uncertainty in their decisions. Unlike diversified investors, managers have undiversified human capital tied to the firm's performance. Therefore, they are particularly sensitive to uncertainty (Panousi and Papanikolaou, 2012). Izhakian and Yermack (2017) find that executives exercise their stock options significantly earlier and accept a significantly lower monetary payout when the firm experiences high Knightian uncertainty. Izhakian et al. (2022) document that managers actively shift financing toward debt and away from equity as Knightian uncertainty rises, consistent with a preference for fixed claims over the valuation of residual equity under Knightian uncertainty.

In preparing financial information, a preparer who is averse to Knightian uncertainty places greater weight on adverse outcomes, resulting in more conservative estimates that guard against unfavorable ex post outcomes (Hansen and Sargent, 2008). This pessimistic belief formation is not intentional manipulation but a mechanism for resolving ambiguity about the underlying probability model.

From a utility perspective, Knightian uncertainty aversion can be fully rational once probabilities are ill-defined (Gilboa and Schmeidler, 1989). Smooth ambiguity preferences capture a distinct attitude toward Knightian uncertainty that is separate from risk (Klibanoff et al., 2005). Under these preferences, pessimism can be interpreted as an optimal response to potential model misspecification (Hansen and Sargent, 2001).

2.2. Knightian uncertainty and fair value estimation

ASC 820 establishes a fair value hierarchy that prioritizes inputs based on their observability (FASB, 2006; Landsman, 2007). Level 1 measurements leave little room for managerial discretion because they are based on quoted prices in active markets. In contrast, Level 2 and

Level 3 measurements require managers to bridge the gap between limited market data and reported value (Barth, 1994; Penman, 2007). For Level 3 items, such as goodwill, indefinite-lived intangibles, and complex derivatives, inputs are unobservable, requiring managers to internally construct the probability distribution of future cash flows and uncertainty premiums (Song et al., 2010; Ramanna and Watts, 2012). Similarly, Level 2 measurements often rely on evaluated pricing services or matrix pricing, which require judgment in selecting and adjusting comparable benchmarks (Cici et al., 2011).

The judgment involved in Level 2 and Level 3 measurement allows Knightian uncertainty aversion to affect the valuation process. When information is imprecise, managers cannot simply calculate a mathematical expectation. Instead, they engage in asymmetric belief formation. Theory suggests that preparers who are averse to Knightian uncertainty place greater weight on adverse scenarios and less weight on favorable ones, effectively embedding a precautionary discount into the reported number (Epstein and Schneider, 2003; Hansen and Sargent, 2008).

This mechanism predicts more pessimistic estimates, in contrast to the opportunistic inflation emphasized by standard agency theory. Prior literature typically characterizes fair value discretion as a tool for opportunistic inflation, whereby managers increase estimates to maximize compensation or avoid covenant violations (Ramanna and Watts, 2012; Hanley et al., 2018). We propose that aversion to Knightian uncertainty is a distinct source of pessimism that may constrain this opportunism. When Knightian uncertainty is greater, aversion to it leads to reported fair value estimates that are lower than a risk-neutral assessment of fundamentals would imply (Brenner and Izhakian, 2018).

On the other hand, real options theory provides a different directional prediction. This theory suggests that when a decision is irreversible, agents have the "option to wait" before committing to the irreversible action (Bernanke, 1983; Bloom, 2009; Arif et al., 2016). The value of the option to wait increases with uncertainty under a classic option-pricing model, increasing the likelihood of waiting under greater economic uncertainty. Although existing studies mainly focus on investment decisions, real options theory can also be applied to Knightian uncertainty and fair value estimation. Recognizing a write-down is a quasi-irreversible accounting decision that normally signals a permanent decline in value. If managers view Knightian uncertainty as temporary noise that may resolve favorably, they may prefer to delay recognition to preserve the asset's option value, engaging in a wait-and-see strategy rather than an immediate write-down (Roychowdhury and Martin, 2013). Greater Knightian uncertainty therefore gives rise to two competing predictions. First, greater Knightian uncertainty makes waiting for new information more valuable, leading managers to delay write-downs. Second, aversion to Knightian uncertainty leads managers to place greater weight on adverse outcomes, producing more pessimistic valuations.

2.3. Hypotheses development

We expect managers to report lower Level 2 and Level 3 estimates when faced with greater Knightian uncertainty. For Level 3 fair value estimates, managers must derive significant unobservable inputs, such as long-term growth rates and specific uncertainty premia, within valuation frameworks, allowing Knightian-uncertainty-driven pessimism to affect parameter selection (Altamuro and Zhang, 2013). For Level 2 measurements, such as mutual fund bond valuations, managers have discretion over whether to accept or adjust evaluated prices. We also expect economic uncertainty to give rise to an option to wait in many Level 3 write-down decisions, such as goodwill impairments, given their quasi-irreversible nature.

2.3.1. Level 3 fair value: goodwill impairment

Under ASC 350 (formerly SFAS 142), goodwill is tested for impairment annually. This test relies almost exclusively on unobservable Level 3 inputs, such as projected future cash flows and discount rates, providing significant scope for managerial discretion to influence the reported value (Ramanna and Watts, 2012). The goodwill impairment test fundamentally involves two distinct economic decisions: the decision to record a write-down (*recognition*) and the determination of the loss magnitude (*measurement*) (Glaum et al., 2018). We posit that the two theoretical forces discussed in Section 2.2 map differently to these two stages.

The first *recognition* decision involves determining whether an impairment exists. Recognizing an impairment is a quasi-irreversible accounting decision: once a write-down is recorded, U.S. GAAP prohibits reversing the charge if the asset's value subsequently recovers. Roychowdhury and Martin (2013) argue that managers may delay recording write-downs to preserve the option value of waiting. They frame such activities as discretionary conservatism. In this context, managers may view different uncertainties regarding future payoffs as temporary noise. To avoid the Type I error of writing down an asset that cannot be recovered, they are incentivized to delay recognition. This logic predicts that high Knightian uncertainty will raise the threshold for recognition.

However, once managers recognize an impairment and forgo the option to wait, they must determine the magnitude of the write-down. This measurement process requires the construction of a specific probability distribution of future cash flows to derive a fair value estimate. In the measurement phase, Knightian uncertainty aversion becomes the dominant force. Unlike the binary recognition decision, measurement is a continuous estimation task where Knightian-uncertainty-averse agents must adopt a belief structure to resolve epistemic uncertainty. Theory posits that such agents systematically overweight adverse probability distributions and underweight favorable ones to insulate themselves against model misspecification (Epstein and Schneider, 2003; Hansen and Sargent, 2008). Therefore, conditional on an impairment being recognized, we expect Knightian uncertainty to result in more pessimistic valuations (i.e., larger impairment charges).

Based on the discussion above, we propose the following hypotheses:

> **H1a:** *Higher Knightian uncertainty is associated with a lower likelihood of recognizing a goodwill impairment.*
>
> ***H1b:*** *Conditional on recognition, higher Knightian uncertainty is associated with a larger magnitude of goodwill impairment.*

2.3.2. Level 2 fair value: mutual fund bond valuation

To examine Level 2 fair value estimates, we focus on mutual fund managers' valuations of corporate bonds. Corporate bonds without active quoted prices are generally classified as Level 2 when their valuations rely principally on observable market inputs (FASB, 2006). These valuations rely on indirectly observable inputs, such as matrix pricing and evaluated pricing services, and require managers to exercise judgment in selecting and adjusting comparable benchmarks (Cici et al., 2011).

The Investment Company Act of 1940 and SEC regulations require mutual funds to disclose their portfolio holdings and fair values on a quarterly or semiannual basis (e.g., via Form N-PORT and Form N-CSR). These fair value measurements must follow U.S. GAAP (ASC 820), requiring managers to assign a specific fair value at each reporting date.

We posit that Knightian uncertainty leads managers to report lower fair values. Knightian uncertainty aversion theory predicts that when probabilities are ambiguous, managers place greater weight on adverse outcomes when selecting valuation inputs (e.g., comparable trades and matrix adjustments), resulting in more pessimistic estimates (Gilboa and Schmeidler, 1989). This pessimistic input selection leads to lower reported values relative to market benchmarks.

However, this prediction is not without theoretical tension. Prior literature suggests that mutual fund flows are sensitive to net asset value volatility, creating incentives for managers to smooth valuation fluctuations. This smoothing can manifest as stale pricing, including a reluctance to fully reflect downward price changes (Cici et al., 2011; Choi et al., 2022). Although these incentives are always present, Knightian uncertainty increases valuation ambiguity by weakening the informativeness of observable inputs and widening the range of plausible estimates. This ambiguity expands the set of defensible valuations and creates greater scope for managers to exercise discretion in ways that stabilize reported values. As a result, managers may report values above current market conditions to preserve portfolio stability.

Taken together, these competing forces yield opposing predictions. Aversion to Knightian uncertainty predicts downward adjustments in reported values, whereas agency-based smoothing incentives predict upward adjustments. Although both channels may operate, we expect the pessimism channel to dominate because valuation inputs directly reflect managers' beliefs about future outcomes. Accordingly, we propose the following hypothesis:

> **H2:** *Higher Knightian uncertainty is associated with lower reported fair values for corporate bonds held by mutual funds.*

3. Data and sample

3.1. Data

3.1.1. Estimating Knightian uncertainty

Prior literature uses several empirical proxies for uncertainty. Return-based measures, including conditional volatility and realized volatility, capture variation in realized outcomes (Andersen et al., 2001; Arif et al., 2016). Implied-volatility indices such as the VIX reflect risk-neutral expected variance and incorporate risk premia and investor risk aversion (Bloom, 2009; Bekaert et al., 2013). Newspaper-based policy uncertainty measures capture uncertainty about regulatory and fiscal frameworks (Baker et al., 2016), while macro-uncertainty measures capture common volatility in the unforecastable components of large panels of indicators (Jurado et al., 2015). Disagreement proxies, such as analyst-forecast dispersion, capture heterogeneity in beliefs across agents (Miller, 1977; Diether et al., 2002). Although informative for many questions, these proxies do not directly measure uncertainty about the probability distribution itself—that is, uncertainty over the probabilities assigned to states (Knight, 1921; Ellsberg, 1961; Halevy, 2007).

To address this limitation, we follow the expected utility with uncertain probabilities (EUUP) framework and the empirical implementation developed by Izhakian (2017, 2020).[2] We use firm-level monthly Knightian uncertainty estimates constructed from intraday TAQ

[2] Knightian uncertainty is defined as $\mho^2 [r_i] \equiv \frac{1}{\sqrt{\omega(1-\omega)}} \sum_{j=1}^{n} E[P_i(B_j)] Var[P_i(B_j)]$. Appendix A and Izhakian (2020) provide additional details.

returns following Izhakian (2017, 2020). The measure captures variation in the probabilities assigned to a firm's future payoffs rather than variation in the payoffs themselves. For each firm-year, we calculate Knightian uncertainty as the mean of the firm's monthly estimates during that fiscal year.

3.1.2. Measuring fair value estimates

We construct two sets of dependent variables to examine the relation between Knightian uncertainty and Level 3 and Level 2 fair value estimates. For Level 3 estimates, we examine both the recognition and measurement of goodwill impairments. We measure recognition using a goodwill impairment indicator that equals one if the firm recognizes a goodwill impairment in year t and zero otherwise. We measure the magnitude of goodwill impairment using the goodwill impairment charge recognized in year t, scaled by total assets at the beginning of the fiscal year. We scale by beginning-of-period total assets because the impairment charge mechanically reduces end-of-period total assets. The resulting measure captures the magnitude of the recognized goodwill write-down relative to firm size.

For Level 2 estimates, we examine mutual fund managers' valuations of inactively traded corporate bonds. When active market quotes are unavailable, managers typically rely on matrix pricing or analyses of comparable securities (Cici et al., 2011). Specifically, we measure the log difference between the fund-reported valuation and a benchmark price. For bond j held by fund k at time t, the markdown is

$$Markdown_{j,k,t} = \ln\left(Price_{j,k,t}^{\text{Fund}}\right) - \ln\left(Price_{j,t}^{\text{Market Benchmark}}\right),$$

where $Price_{j,t}^{\text{Market Benchmark}}$ is either (a) the evaluated price supplied by Bloomberg's evaluated pricing service (BVAL) for bond j on reporting date t or (b) the midpoint of the buy-side and sell-side volume-weighted average prices on the bond's last trading day in the reporting month.

3.1.3. Control variables

For the Level 3 tests, we control for fundamental economic deterioration and investment opportunities. These controls include firm size (*Size*), return on assets (*ROA*), annual stock returns (*Return*), leverage (*Leverage*), Altman Z-score (*AltmanZ*), book-to-market ratio (*Book-to-Market*), and sales growth (*Sales Growth*). We also control for capital expenditures (*CAPX*) to account for a real-options channel through which uncertainty may affect the asset base. To control for strategic reporting incentives, we include a loss indicator variable (*Loss*). We use institutional ownership (*Institutional Holding*) as a proxy for external monitoring pressure.

In addition, greater outcome risk can increase discount rates and thereby reduce fair values independently of Knightian uncertainty aversion. We control for risk using two proxies: the fiscal-year average of the firm's monthly estimates of average daily return variance (*Risk*) and operating cash flow volatility (*CFO Volatility*). We also control for analyst coverage (*Analyst Forecasts*) and forecast dispersion (*Forecast Dispersion*) to mitigate the concern that our results are driven by information asymmetry or disagreement among market participants rather than Knightian uncertainty.

For the mutual fund analysis, we control for bond-specific characteristics associated with valuation complexity and market liquidity. We include bond age (*Bond Age*), time to maturity (*Maturity*), coupon rate (*Coupon*), and high-yield status (*High Yield*). We also include restricted-security status (*Restricted*), Amihud illiquidity (*Amihud*), *Order Imbalance*, *Days since Last Trade*, and *Days with Trades* to control for liquidity premiums and market microstructure frictions. We include accrued interest (*Interest Accruals*) to control for its potential effect on bond valuations. We also include issuer-level characteristics to control for fundamentals, such as firm size (*Size*), leverage (*Leverage*), financial distress (*AltmanZ*), and stock performance (*Return*). Finally, we include fund-quarter fixed effects to absorb fund-level factors common to the valuation of all holdings in a quarter, including valuation policies and potential strategic incentives, and control for each position's relative importance in the fund's portfolio (*Portfolio Weight*).

3.2. Sample descriptives

We construct two samples to test our hypotheses: a firm-year panel of goodwill impairments for the Level 3 tests (Level 3 sample) and a mutual-fund–bond–quarter panel of mutual funds' corporate bond valuations for the Level 2 tests (Level 2 sample).

Our Level 3 sample begins with all U.S. public firms covered by Compustat and CRSP with available data to calculate Knightian uncertainty using intraday TAQ data. We include firms with a positive goodwill balance or a goodwill acquisition in at least one fiscal year. We also require firms to have non-missing control variables. To mitigate the influence of outliers, we winsorize all continuous variables at the 1st and 99th percentiles. The final Level 3 sample consists of 47,585 firm-year observations from 1998 to 2024. This count, and the observation counts in Table 1, are descriptive-sample counts; the estimation samples in the goodwill impairment tests are smaller—47,420 firm-years in Tables 2 and 3 and 42,813 in Table 7—because those specifications require the corrected total shareholder return and common support across the return measures.

For the Level 2 sample, we obtain all quarterly Form N-PORT filings from the SEC covering 2019 Q4 through 2024 Q4. We identify all corporate bond holdings that mutual funds classify as Level 2. We use Form N-PORT rather than Morningstar or CRSP Mutual Fund data because the commercial databases combine mandated and voluntary disclosures and provide holdings at inconsistent frequencies (Elton et al., 2010; Schwarz and Potter, 2016). We then merge the holdings and valuation data with bond characteristics from Mergent FISD and transaction data from TRACE. We focus on infrequently traded corporate bonds, for which managers rely more heavily on indirectly observable inputs and judgment. Specifically, we require that a bond did not trade on the reporting date and that its most recent institutional trade occurred more than five calendar days earlier. We obtain quarterly financial data for the bond issuers from Compustat. All continuous variables are winsorized at the 1st and 99th percentiles. Our final Level 2 sample includes 142,508 fund–bond–quarter observations.

Table 1 reports the descriptive statistics for the variables used in our analysis. For the Level 3 sample, our key independent variable, *Knightian Uncertainty*, has a mean of 0.050 and a median of 0.033. The standard deviation of 0.050 indicates substantial cross-sectional variation. For the dependent variable, the mean *Goodwill Impairment* is 0.005, indicating an average annual goodwill impairment of 0.5% of total assets. The *Goodwill Impairment Indicator* has a mean of 0.109, indicating that approximately 10.9% of the firm-year observations include a recognized goodwill impairment.

Regarding firm characteristics, the mean and median of firm size are 7.547 and 7.488, respectively. The leverage ratio has a mean of 0.569. The average firm has an ROA of 0.017 and a book-to-market ratio of 0.560.

For bond issuers in the Level 2 sample, the mean of *Knightian Uncertainty* is 0.105. The means of *Markdown_Evaluated Price* and *Markdown_Recent Price* are both −0.003, and their medians are −0.002 and −0.001, respectively. This evidence suggests that, on average, fund-reported valuations are slightly below benchmark prices. In terms of issuer characteristics, firms in the bond sample are generally larger (mean *Size* of 10.348) than those in the Level 3 sample. The sample's high-yield share, time to maturity, and time since the last trade are consistent with those reported by Cici et al. (2011).

3.3. Inference

Several of our tests use annual firm-level panels in which fair value decisions cluster in calendar time. Because such clustering can induce contemporaneous correlation across firms, all standard errors, *t*-statistics, and significance stars reported in the goodwill impairment tests (Tables 2, 3, and 7) are based on two-way clustering by firm and calendar year, holding the estimation sample, variable construction, fixed effects, and estimator fixed so that the point estimates are unchanged and only the variance estimator differs. Because the calendar-year dimension contains only 27 clusters, we additionally report, for the focal coefficients, wild cluster bootstrap p-values that impose the null (MacKinnon et al., 2021), using 9,999 replications under both Rademacher and Webb weights and resampling on the calendar-year dimension, implemented through `boottest` (Roodman et al., 2019). We flag any coefficient whose significance class differs between the analytic two-way result and the bootstrap, and we treat p-values near conventional thresholds as borderline rather than as decisive.

4. Knightian uncertainty and Level 3 fair value estimates

4.1. Baseline results

We examine whether Knightian uncertainty is associated with managers' Level 3 fair value estimates in the goodwill impairment setting. We first examine the unconditional relation between Knightian uncertainty and goodwill impairment charges. Table 2 reports the baseline regressions of *Goodwill Impairment* on *Knightian Uncertainty*, controlling for expected risk and a comprehensive set of firm characteristics.

The coefficient on *Knightian Uncertainty* is positive across specifications using different combinations of year, industry, and firm fixed effects, and its sign and magnitude are stable under alternative variance estimators. Its statistical significance, however, is sensitive to how we model dependence across firms in calendar time. Under standard errors clustered by firm, the coefficient is significant in Columns (1) and (3); once we admit contemporaneous correlation across firms by clustering on both firm and calendar year, it is no longer statistically distinguishable from zero (Column (1), $p = 0.41$; Column (3), $p = 0.24$). In our most restrictive specification, which includes firm and year fixed effects, the coefficient is 0.0131 ($t = 1.20$ under two-way clustering). The wild cluster bootstrap agrees (p = 0.70, 0.90, and 0.40 in Columns (1)–(3)). We therefore read Table 2 as descriptive of an unconditional association whose sign is stable but whose statistical reliability is sensitive to the treatment of calendar-time dependence, and we base our inferences about the recognition and measurement margins on the hurdle model in Table 3.

The unconditional association is also concentrated in a single year. An exact Frisch–Waugh–Lovell decomposition attributes a majority of the Table 2 numerator to calendar-year 2008, which accounts for roughly 4.5% of the sample. The concentration is broad rather than the product of a few observations: it spans more than two thousand firms, no single observation contributes more than about 3% of the numerator, and the sample contains no duplicated observations. The 2008 observations are genuine and economically important—the year contains the largest goodwill impairments in the sample—so we retain them in all specifications. As a sensitivity check, excluding 2008 reverses the sign of the coefficient in Columns (1) and (2) and reduces it by roughly ninety percent in Column (3), shifts of between 2.7 and 3.3 standard errors, whereas no other single year moves any estimate by more than half a standard error. We report this concentration as a description of the sources of the unconditional association.

4.2. Recognition versus measurement: a two-stage hurdle model

As discussed in Section 2.3.1, goodwill impairment includes two distinct economic decisions: (1) whether to recognize an impairment (recognition, the extensive margin), and (2) conditional on recognition, how large the impairment is (measurement, the intensive margin). Knightian uncertainty has two countervailing effects on these decisions. Greater Knightian uncertainty increases the value of waiting before recognizing a quasi-irreversible impairment. Conditional on recognition, however, managers averse to Knightian uncertainty place greater weight on adverse states when measuring the impairment. To separately identify these two margins, we follow the existing literature (e.g., Cragg, 1971; Beatty and Weber, 2006; Nessa, 2017) and estimate a two-stage hurdle model. Table 3 reports the results. In the first stage, we examine the probability of recording any impairment (*Goodwill Impairment Indicator*) using a linear probability model with fixed effects. In the second stage, we use OLS to examine the association between Knightian uncertainty and the magnitude of goodwill impairments among firm-years with a positive impairment. We measure the magnitude of the impairment (*Goodwill Impairment*) as the recognized charge divided by total assets at the beginning of the fiscal year.

In the specifications that include firm fixed effects, higher Knightian uncertainty is associated with a lower likelihood of impairment recognition. The coefficient on *Knightian*

Uncertainty is -0.1844 ($t = -2.53$) with firm and year fixed effects (Column (3)), and it remains significant at the 5% level under two-way clustering by firm and calendar year and under the wild cluster bootstrap. In contrast, the second-stage results indicate that, conditional on recognizing an impairment, Knightian uncertainty is associated with larger impairment magnitudes. The coefficient on *Knightian Uncertainty* is 0.0583 ($t = 2.21$) with firm and year fixed effects (Column (6)); this estimate is significant at the 5% level under two-way clustering but is borderline at the 10% level under the wild cluster bootstrap, whereas the conditional-magnitude estimates in Columns (4) and (5) are more robust across inference methods.

Taken together, the results documented in Table 3 suggest that managers facing greater Knightian uncertainty are more likely to delay impairment recognition but record larger goodwill impairments conditional on recognition. This pattern is consistent with theory suggesting that managers facing greater Knightian uncertainty preserve the option to wait by delaying recognition. Conditional on recognition, however, managers averse to Knightian uncertainty form more pessimistic cash flow forecasts and therefore record larger impairments. The positive unconditional association in Table 2 is therefore consistent with larger impairments conditional on recognition outweighing the lower likelihood of recognition.

To investigate potential selection concerns, Appendix C reports a Heckman two-step selection model with year fixed effects, where impairment recognition is modeled in a probit first stage and impairment magnitude is modeled in the second stage, using an indicator for whether the firm's book-to-market ratio exceeds 1.0 (*Book Greater than Market*) as a plausible exclusion restriction in the selection equation.[3] The selection-equation coefficient

[3] We define *Book Greater than Market* as an indicator equal to one when the firm's book-to-market ratio exceeds 1.0. This condition identifies firms for which our measure of book value exceeds market value, providing market-based evidence that the fair value of one or more reporting units may be below their carrying amounts. Under ASC 350, firms that perform the qualitative assessment consider the totality of relevant events and circumstances, including a sustained decline in share price, when deciding whether to proceed to quantitative impairment testing. We use the indicator as an exclusion restriction because it should primarily predict whether a firm proceeds toward impairment recognition. Conditional on recognition and the continuous economic fundamentals included in the outcome equation, we assume that crossing this threshold does not independently determine the magnitude of the impairment charge.

on Knightian uncertainty remains negative but is not statistically significant. The outcome-equation coefficient remains positive and statistically significant. The positive inverse Mills ratio is statistically significant at the 5% level, providing evidence of nonrandom selection into impairment recognition.

4.3. A shock to managerial discretion

We next provide corroborating evidence that Knightian uncertainty affects managers' fair value estimates by examining ASU 2011-08, an externally imposed change that increased the role of managers' judgment in goodwill impairment testing. Before ASU 2011-08, managers could, under limited circumstances, carry forward a prior-period detailed fair value calculation. The standard eliminated that option and introduced an optional qualitative screen ("Step 0"). Managers may elect the screen and avoid a quantitative test if they conclude that it is not more likely than not that a reporting unit's fair value is below its carrying amount. Managers who conclude that the threshold is met—or who bypass the qualitative screen—must perform the quantitative test. Thus, ASU 2011-08 required a current qualitative or quantitative assessment without changing the quantitative impairment test itself (FASB, 2011).[4] Because ASU 2011-08 operates through the qualitative assessment that determines whether the quantitative impairment test is performed, its most direct effect is on the recognition decision. We therefore examine goodwill impairment recognition as the primary outcome and ask whether the relation between Knightian uncertainty and recognition changed

[4]We considered ASU 2017-04 as an additional setting in which managers' discretion over goodwill impairment changed. Before ASU 2017-04, a reporting unit that failed Step 1 proceeded to Step 2, but failing Step 1 did not itself require a goodwill impairment. In Step 2, managers hypothetically assigned the reporting unit's fair value to its identifiable assets and liabilities and treated the residual as the implied fair value of goodwill. This allocation did not change the recorded amounts of the individual assets and liabilities. Knightian uncertainty aversion could have opposing effects within this calculation: a lower reporting-unit fair value would reduce implied goodwill, whereas lower values assigned to identifiable net assets would increase the residual attributed to goodwill. ASU 2017-04 eliminated Step 2 and instead measures impairment directly as the amount by which the reporting unit's carrying amount exceeds its fair value, limited to recorded goodwill (FASB, 2017). Because the standard both removed the allocation judgment and changed how reporting-unit fair value maps into the impairment charge, we cannot form an unambiguous prediction about its effect on the relation between Knightian uncertainty and goodwill impairments. We therefore do not use ASU 2017-04 as a corroborating test. Its effective date for SEC filers also substantially overlaps the COVID-19 pandemic and related economic shocks.

after the standard, and whether that change was stronger among firms with more goodwill at stake before implementation.

Table 4 presents the results. We estimate a linear-probability model of goodwill impairment recognition on the risk set of firm-years with positive beginning goodwill, and we interact Knightian uncertainty measured before the fiscal-period end with an indicator for the post-adoption period (*Post*) and with a firm's predetermined goodwill exposure, measured as the standardized average ratio of beginning goodwill to beginning assets over the two pre-adoption fiscal years. The model includes firm and decision-month fixed effects and is estimated over a balanced window of the fiscal years surrounding adoption. Because implementation is common to all firms rather than staggered across treated and untreated units, we interpret the design as an associative interrupted-time-series mechanism test with exposure heterogeneity rather than as a causal difference-in-differences design. In the primary two-period window (Column (2)), the coefficient on the triple interaction is -0.4699 (two-way clustered $t = -3.84$), and it is similar across the one-, three-, and four-period windows.

The triple interaction describes a gradient across predetermined exposure that crosses from positive at low exposure to negative at high exposure. A one-standard-deviation increase in Knightian uncertainty is associated with a 2.84-percentage-point higher post-versus-pre change in recognition at the 25th percentile of exposure but a 3.50-percentage-point lower change at the 90th percentile. The negative association is therefore concentrated among firms with the greatest predetermined goodwill exposure, consistent with uncertainty-averse managers becoming more likely to defer recognition when they have more goodwill at stake and greater discretion over the assessment. The change in the recognition margin does not extend to the conditional magnitude of impairments: among the firm-years in which an impairment is recognized, the analogous triple interaction is small and statistically insignificant (0.0488, $p = 0.60$), so we do not interpret the standard as changing the amount recognized conditional on recognition. A joint test of the pre-adoption exposure-specific Knightian-uncertainty coefficients does not reject the absence of differential pre-trends ($p = 0.67$), and the focal estimate is robust to alternative windows, to estimation on maximum common support, and to a wild cluster bootstrap. Because this test and the unconditional association in Table 2 estimate different quantities—the change in recognition probability across predetermined exposure rather than the level of the unconditional impairment amount—we do not compare their coefficients directly.

5. Knightian uncertainty and Level 2 fair value estimates

We next test H2 using mutual fund managers' valuations of corporate bond holdings. Under the Investment Company Act of 1940 and subsequent SEC regulations, mutual funds are required to disclose their portfolio holdings and fair value measurements periodically. Prior to 2004, mutual funds were required to file these reports with the SEC semiannually via Form N-30D. The frequency of the reports switched to quarterly in 2004 with the introduction of Form N-Q. In the late 2010s, the SEC replaced Form N-Q with Form N-PORT, which provides machine-readable data on corporate bond valuations reported by mutual fund managers.[5]

Many corporate bonds do not trade actively. For these bonds, fair value inputs are only indirectly observable on the reporting date. Managers must therefore exercise judgment when using pricing models and observable inputs from comparable securities to estimate fair value (Cici et al., 2011; Choi et al., 2022). We focus on bonds that did not trade on the quarterly reporting date and whose most recent trade occurred more than five calendar days earlier.

We construct two markdown measures that compare the fund-reported valuation with a market benchmark. *Markdown_Evaluated Price* is the log difference between the fund-reported valuation and Bloomberg's evaluated price (BVAL)[6] for the same bond on the reporting date. *Markdown_Recent Price* is the log difference between the fund-reported valuation and a recent-price benchmark. We measure the recent-price benchmark as the

[5]We use Form N-PORT rather than Morningstar or CRSP Mutual Fund data because the commercial databases combine mandated and voluntary disclosures and provide holdings at inconsistent frequencies (Elton et al., 2010; Schwarz and Potter, 2016).

[6]Bloomberg evaluated prices are widely used as pricing references in fixed-income markets. Other service providers include Refinitiv (LSEG) and ICE Data Services.

midpoint of the buy-side and sell-side volume-weighted average prices on the bond's last trading day in the reporting month. A lower value indicates a valuation that is more pessimistic relative to the benchmark market price. We include fund-quarter fixed effects to absorb fund-level factors common to the valuation of all holdings in a quarter, including valuation policies and potential strategic incentives. Table 5 presents the OLS estimates. The results are consistent across specifications that use different fixed-effect structures. Across these specifications, greater issuer-level Knightian uncertainty is associated with lower fund-reported fair values relative to benchmark prices. Specifically, the estimated coefficients in Columns (2) and (5) indicate that a one-standard-deviation increase in Knightian uncertainty is associated with additional markdowns of approximately 0.023% relative to Bloomberg's evaluated price and 0.031% relative to the recent-price benchmark, respectively.

A potential concern is that recent transaction prices may reflect liquidity conditions associated with Knightian uncertainty, making it difficult to distinguish fund managers' valuations from movements in the benchmark. To mitigate this concern, we estimate a latent-price benchmark using the prices, coupon rates, high-yield status, maturities, and ages of liquid bonds from the same issuer-month.[7] The resulting synthetic benchmark relies on observable inputs from liquid bonds issued by the same firm rather than on the potentially illiquid transaction price of the bond being valued (Choi et al., 2022).

Table 6 reports the estimates using *Markdown_Latent Price* as the dependent variable.[8]

[7]We classify a bond as liquid if it trades on the reporting date or its most recent trade occurred no more than five calendar days earlier. We estimate the latent-price model only for issuer-months with at least 20 liquid bonds and apply it only within the maturity range represented by those bonds.

[8]As a robustness check on the two-step latent-price design, we bootstrap the Knightian-uncertainty coefficients with a nonparametric issuer-cluster bootstrap that resamples the 47 issuers contributing to the second-stage regressions and, within each resample, *re-estimates both the latent-price first stage and the second stage* (5,000 replications, seed 12345, single continuous random-number stream; source-correct quarterly return). Based on the bootstrap standard errors, the Knightian-uncertainty coefficient is significant at the 10% level in column (1) ($p = 0.085$) and is marginally *above* the 10% level in column (2) ($p = 0.103$). The result is stable across alternative resampling choices: retaining eleven issuers that enter only the first stage (a literal-union universe) gives $p = 0.081$ and $p = 0.099$, and holding the estimated latent-price inputs fixed rather than re-estimating them gives $p = 0.083$ and $p = 0.100$; in every variant column (1) is significant at 10% and column (2) sits within a whisker of the 10% threshold. This untabulated bootstrap is more conservative than, and does not alter, the two-way-clustered analytic inference reported in the table.

The negative association between Knightian uncertainty and valuation markdown persists and strengthens under the latent benchmark. The coefficient in Column (1) indicates that a one-standard-deviation increase in Knightian uncertainty is associated with an additional markdown of approximately 0.48% relative to the latent benchmark, using the Level 2 sample standard deviation of Knightian uncertainty (0.068; Table 1).

6. Economic consequences

Our results are consistent with Knightian uncertainty affecting managers' fair value estimates. An important remaining question is whether investors respond differently to the more pessimistic estimates associated with Knightian uncertainty. We therefore examine two economic consequences. First, we examine whether stock-market reactions to goodwill impairments vary with Knightian uncertainty. Second, we examine whether subsequent mutual fund flows vary when bond markdowns are more strongly associated with Knightian uncertainty.

6.1. Market reaction to goodwill impairments

We first examine whether the market response to goodwill impairments varies with Knightian uncertainty. If managers' goodwill estimates are more pessimistic when Knightian uncertainty is greater, a given impairment amount may convey less incremental information about underlying cash flows. We therefore predict a weaker market response to a given goodwill impairment amount when Knightian uncertainty is greater.

Table 7 reports the results. The dependent variable is the absolute cumulative abnormal return around the earnings announcement, *abs(CAR(−2,+2))*, which captures the magnitude of the market's response to the information released at the announcement. Panel A focuses on the impairment recognition decision, while Panel B focuses on the impairment magnitude decision. In both panels, we interact the impairment variable with Knightian uncertainty to test whether the market reaction to impairment-related news differs when the firm faces greater Knightian uncertainty.

Panel A provides some evidence on the recognition margin. Although the interaction between *Goodwill Impairment Indicator* and *Knightian Uncertainty* is negative and statistically significant at the 5% level in the baseline specification, the estimate becomes statistically insignificant once additional controls and interaction terms are introduced. We therefore do not view the recognition margin evidence as robust.

Among firm-years in which a goodwill impairment is recognized, the interaction between *Goodwill Impairment* and *Knightian Uncertainty* is negative in all three specifications. Under two-way clustering, the association is significant at the 5% level or better in Columns (1) and (3), including at the 1% level in Column (3); Column (2) is borderline at the 10% level. These results are consistent with the market reacting less strongly to a given impairment amount when Knightian uncertainty is greater. This pattern suggests that investors place less weight on impairment amounts when Knightian uncertainty leads managers to make more pessimistic fair value estimates.

6.2. Fund flow response to corporate bond markdowns

We next examine whether the bond-level markdown pattern documented in Section 5 is associated with subsequent fund flows. In those tests, mutual fund managers report lower valuations relative to benchmark prices for bonds issued by firms with greater Knightian uncertainty. We predict weaker subsequent flows when a fund's markdowns are concentrated in these bonds.

Table 8 reports the results. The dependent variables are subsequent net flows (*Future Net Flow*), inflows (*Future Inflow*), and outflows (*Future Outflow*). The main independent variable pairs each bond's markdown with its issuer's Knightian uncertainty and then aggregates these products using portfolio weights (*Aggregated Markdowns* × *Knightian Uncertainty*).[9]

[9] We construct the interaction term at the individual bond level before aggregating it to the fund-quarter level so that each bond's markdown remains paired with its issuer's Knightian uncertainty. Specifically, before scaling, we calculate $\sum_b \omega_{bft}(Markdown_{bft} \times Knightian\ Uncertainty_{bft})$, where ω_{bft} is the bond's portfolio weight. We multiply this portfolio-weighted sum by $-1{,}000$ for ease of interpretation. This construction distinguishes between funds with the same average markdown and average Knightian uncertainty but different concentrations of larger markdowns among bonds issued by firms with greater Knightian uncertainty.

Panel A reports fund flows for the first month reported in the fund's next Form N-PORT filing, while Panel B reports flows for all three months reported in that filing. Fund flows are scaled by the fund's current net assets, taken from the most recent amended Form N-PORT filing. The Panel B flow measures are winsorized at the 1st and 99th percentiles, and the Panel B sample comprises the 17,214 fund-quarters with an adjacent next-quarter filing.

Across the one-month and one-quarter horizons, the estimated coefficient on *Aggregated Markdowns* $\times$ *Knightian Uncertainty* is negative and statistically significant for *Future Net Flow* and *Future Inflow*, but not for *Future Outflow*. These results suggest that funds whose markdowns are concentrated in bonds issued by firms with greater Knightian uncertainty subsequently receive lower net flows and inflows.

Taken together, these results suggest that the stock market reacts less strongly to a given goodwill impairment amount when Knightian uncertainty is greater. Funds also receive lower subsequent net flows and inflows when their markdowns are concentrated in bonds issued by firms with greater Knightian uncertainty.

A natural question is why managers continue to form more pessimistic fair value estimates if investors respond less to them. These responses can coexist without implying that either managers or investors behave irrationally. When probabilities are ill-defined, aversion to Knightian uncertainty can be rational (Klibanoff et al., 2005; Epstein and Schneider, 2010). Managers may therefore form more pessimistic estimates as a rational response to probability-model uncertainty, while investors may rationally place less weight on those estimates when allocating capital. Although we do not test the mechanisms underlying either response, this distinction explains why a weaker investor response need not eliminate more pessimistic estimation under Knightian uncertainty.

7. Conclusion

Fair value estimates play a central role in financial reporting (Barth, 2006), yet little is known about how managers form these estimates when the probability distribution governing future outcomes is unknown. This question is important because fair value estimation often requires managers to assign weights to alternative scenarios even when their probabilities are not uniquely specified. We examine this question using a recently developed measure of Knightian uncertainty derived from intraday TAQ data in two settings that require substantial judgment: goodwill impairments and mutual funds' valuations of corporate bonds.

We provide evidence that Knightian uncertainty is associated with both Level 3 goodwill impairments and Level 2 corporate bond valuations. In the goodwill setting, managers are less likely to recognize an impairment when Knightian uncertainty is greater but record larger impairment charges conditional on recognition. These results are consistent with managers preserving the option to wait before recognizing a quasi-irreversible impairment and forming more pessimistic fair value estimates once they recognize one. The positive unconditional association is consistent with larger conditional impairments outweighing the lower likelihood of recognition. In the bond setting, mutual fund managers report lower valuations relative to benchmark prices for bonds issued by firms with greater Knightian uncertainty.

We also provide evidence that investor responses vary with Knightian uncertainty. The stock market reacts less strongly to a given goodwill impairment amount when Knightian uncertainty is greater, and funds receive lower subsequent net flows and inflows when their markdowns are concentrated in bonds issued by firms with greater Knightian uncertainty. Taken together, these findings suggest that Knightian uncertainty is associated not only with managers' fair value estimates but also with investors' responses to them.

Overall, our findings are consistent with Knightian uncertainty affecting fair value reporting in two countervailing ways. First, managers are less likely to recognize goodwill impairments when firm-level Knightian uncertainty is greater, a pattern consistent with preserving the option to wait. Second, conditional on recognition, managers record larger goodwill impairment charges when firm-level Knightian uncertainty is greater, a pattern consistent with placing greater weight on adverse outcomes. We find a similar valuation pattern in corporate bonds: mutual fund managers report lower valuations relative to benchmark prices when issuer-level Knightian uncertainty is greater. This evidence is relevant to standard setters and regulators because it suggests that neutrality may depend on the information and verification environment surrounding fair value estimates.

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Table 1: Summary Statistics

Panel A: Level 3 Sample

Variable	N	Mean	P25	P50	P75	SD	Min	Max
Knightian Uncertainty	47,585	0.050	0.014	0.033	0.067	0.050	0.001	0.227
Risk	47,585	0.050	0.007	0.015	0.038	0.108	0.002	1.318
Goodwill Impairment	47,585	0.005	0.000	0.000	0.000	0.024	0.000	0.192
Goodwill Impairment Indicator	47,585	0.109	0.000	0.000	0.000	0.311	0.000	1.000
Size	47,585	7.547	6.229	7.488	8.781	1.869	2.338	12.302
Leverage	47,585	0.569	0.386	0.563	0.745	0.255	0.066	1.377
Book-to-Market	47,585	0.560	0.257	0.465	0.759	0.491	−0.810	3.146
AltmanZ	47,585	4.046	1.380	3.003	5.042	5.349	−13.713	37.156
ROA	47,585	0.017	0.003	0.036	0.081	0.147	−1.017	0.300
Return	47,420	0.138	−0.170	0.079	0.344	0.519	−0.809	2.408
Sales Growth	47,585	0.124	−0.013	0.075	0.190	0.337	−0.664	2.692
Loss	47,585	0.237	0.000	0.000	0.000	0.425	0.000	1.000
CAPX	47,585	3.723	2.242	3.624	5.071	2.013	0.000	8.530
Goodwill Acquisition	47,585	0.154	0.000	0.000	0.000	0.361	0.000	1.000
CFO Volatility	47,585	0.050	0.018	0.034	0.062	0.053	0.002	0.376
Analyst Forecasts	47,585	3.589	2.996	3.555	4.190	0.816	1.386	5.293
Forecast Dispersion	47,585	0.580	0.033	0.087	0.244	2.819	0.002	28.435
Institutional Holding	47,585	0.698	0.543	0.765	0.903	0.274	0.002	1.126
Big 4	47,585	0.756	1.000	1.000	1.000	0.429	0.000	1.000
Going Concern	47,585	0.009	0.000	0.000	0.000	0.097	0.000	1.000
abs(CAR(−2,+2))	42,957	0.068	0.022	0.048	0.093	0.065	0.000	0.297
abs(UE)	42,957	0.010	0.001	0.002	0.006	0.028	0.000	0.220

Table 1: Summary Statistics (Cont.)

Panel B: Level 2 Sample

Variable	N	Mean	P25	P50	P75	SD	Min	Max
Knightian Uncertainty	142,508	0.105	0.056	0.093	0.141	0.068	0.007	0.347
Risk	142,508	0.007	0.003	0.004	0.007	0.012	0.001	0.081
Markdown_Evaluated Price	141,195	−0.003	−0.005	−0.002	0.000	0.004	−0.015	0.006
Markdown_Recent Price	142,508	−0.003	−0.007	−0.001	0.003	0.009	−0.021	0.013
Markdown_Latent Price	20,622	−0.007	−0.034	0.000	0.027	0.065	−0.870	0.239
Size	142,508	10.348	9.423	10.326	11.118	1.243	7.763	14.714
Leverage	142,508	0.709	0.598	0.702	0.805	0.168	0.309	1.290
AltmanZ	142,508	2.785	1.164	2.611	3.928	1.825	0.109	10.584
Return	142,508	0.037	−0.049	0.032	0.119	0.150	−0.469	0.576
Portfolio Weight	142,508	0.062	0.003	0.011	0.042	0.151	0.001	1.125
Restricted	142,508	0.003	0.000	0.000	0.000	0.054	0.000	1.000
Amihud	142,508	0.025	0.005	0.014	0.034	0.028	0.000	0.096
Order Imbalance	142,508	0.002	−0.186	0.000	0.187	0.305	−0.567	0.600
Days since Last Trade	142,508	2.495	2.079	2.398	2.708	0.489	1.946	4.511
Days with Trades	142,508	1.817	1.386	1.792	2.197	0.514	1.099	2.996
Coupon	142,508	4.287	3.375	4.125	4.950	1.396	0.250	13.000
High Yield	142,508	0.377	0.000	0.000	1.000	0.485	0.000	1.000
Bond Age	142,508	5.907	2.658	4.928	8.030	4.272	0.047	16.964
Maturity	142,508	12.042	3.786	8.167	20.832	9.575	0.457	38.954
Interest Accruals	142,508	1.067	0.482	0.983	1.520	0.713	0.024	2.975
Future Net Flow_next month	16,487	0.007	−0.008	0.000	0.014	0.045	−0.135	0.252
Future Inflow_next month	16,487	0.031	0.005	0.017	0.036	0.048	0.000	0.328
Future Outflow_next month	16,487	0.024	0.006	0.015	0.028	0.033	0.000	0.220
Future Net Flow_next quarter	16,473	0.022	−0.025	0.000	0.043	0.121	−0.287	0.693
Future Inflow_next quarter	16,473	0.099	0.025	0.061	0.116	0.138	0.000	0.923
Future Outflow_next quarter	16,473	0.076	0.028	0.053	0.094	0.086	0.000	0.548
Aggregated Markdowns	16,487	0.011	0.000	0.003	0.012	0.024	−0.029	0.136

Table 1: Summary Statistics (Cont.)

Panel B: Level 2 Sample

Variable	N	Mean	P25	P50	P75	SD	Min	Max
Aggregated Markdowns × Knightian Uncertainty	16,487	0.001	0.000	0.000	0.001	0.002	−0.002	0.013
Aggregated Knightian Uncertainty	16,487	0.000	0.000	0.000	0.000	0.001	0.000	0.004
Aggregated Markdowns × Risk	16,487	0.000	0.000	0.000	0.000	0.000	0.000	0.003
Aggregated Risk	16,487	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Fund Performance	16,487	0.003	−0.015	0.001	0.024	0.043	−0.130	0.127
Fund Size	16,487	20.447	19.156	20.428	21.672	1.885	15.956	24.974
Short-Term Leverage	16,487	0.009	0.000	0.000	0.000	0.050	0.000	0.382
Long-Term Leverage	16,487	0.001	0.000	0.000	0.000	0.011	0.000	0.108

Notes: This table reports summary statistics for the variables used in the analyses. The Level 3 sample includes the firm-year variables used in the goodwill impairment analyses. The Level 2 sample includes the issuer-, bond-, holding-, liquidity-, and fund-level variables used in the corporate bond valuation and fund-flow analyses. The number of observations varies across variables because the analyses use different samples and require different data. The observation counts reported here are descriptive-sample counts, not the corrected estimation samples; the regression estimation samples are smaller where the specifications require the corrected total shareholder return and common support across measures (for example, 47,420 firm-years in Tables 2 and 3 and 42,813 in Table 7, versus the 47,585 and 42,957 firm-years described here). Appendix B contains all variable definitions and measurement details.

Table 2: Knightian Uncertainty and Goodwill Impairments

	Dependent Variable: Goodwill Impairment		
	(1)	(2)	(3)
Knightian Uncertainty	**0.0068**	**0.0049**	**0.0131**
	(**0.85**)	(**0.58**)	(**1.20**)
Risk	−0.0039*	−0.0039*	−0.0049*
	(−1.74)	(−1.90)	(−1.77)
Size	−0.0009**	−0.0002	−0.0005
	(−2.48)	(−0.54)	(−0.76)
Leverage	0.0034*	0.0046**	0.0039*
	(1.95)	(2.36)	(1.76)
Book-to-Market	0.0044***	0.0049***	0.0048***
	(2.95)	(3.20)	(3.19)
AltmanZ	0.0005***	0.0005***	0.0006***
	(3.60)	(3.81)	(4.44)
ROA	−0.0529***	−0.0572***	−0.0891***
	(−5.01)	(−5.34)	(−7.12)
Return	−0.0031***	−0.0029***	−0.0018***
	(−6.09)	(−5.82)	(−5.43)
Sales Growth	−0.0022***	−0.0016***	0.0012*
	(−5.83)	(−3.97)	(1.93)
Loss	0.0069***	0.0062***	0.0060***
	(5.07)	(4.76)	(4.70)
CAPX	0.0009***	0.0006***	0.0015***
	(4.64)	(3.69)	(6.29)
Goodwill Acquisition	0.0012***	0.0004	−0.0008**
	(3.14)	(1.23)	(−2.10)
CFO Volatility	−0.0352***	−0.0307***	−0.0042
	(−4.07)	(−3.88)	(−0.69)
Analyst Forecasts	0.0004	0.0004	0.0010*
	(1.07)	(1.05)	(1.97)
Forecast Dispersion	−0.0001	0.0000	0.0007***
	(−0.55)	(−0.39)	(4.40)
Institutional Holding	0.0030***	0.0027***	0.0009
	(3.16)	(2.86)	(0.81)
Big 4	0.0007*	0.0000	0.0012**
	(2.00)	(−0.02)	(2.64)
Going Concern	−0.0055*	−0.0052*	−0.0025
	(−1.79)	(−1.74)	(−0.78)

Table 2: Knightian Uncertainty and Goodwill Impairments (Cont.)

	Dependent Variable: Goodwill Impairment		
	(1)	(2)	(3)
Year FE	Yes	Yes	Yes
Industry FE	No	Yes	No
Firm FE	No	No	Yes
Obs.	47,420	47,420	47,420
Adj. R^2	0.184	0.198	0.321

Notes: This table reports coefficient estimates from OLS regressions examining the relation between Knightian uncertainty and Level 3 fair value estimates, using goodwill impairments as the setting. The dependent variable (*Goodwill Impairment*) equals the goodwill impairment charge recognized during the fiscal year divided by lagged total assets. The main independent variable of interest is *Knightian Uncertainty.* Control variables include *Risk, Size, Leverage, Book-to-Market, AltmanZ, ROA, Return, Sales Growth, Loss, CAPX, Goodwill Acquisition, CFO Volatility, Analyst Forecasts, Forecast Dispersion, Institutional Holding, Big 4*, and *Going Concern.* Appendix B contains all variable definitions. All specifications include year fixed effects; Column (2) also includes industry fixed effects, and Column (3) also includes firm fixed effects. Because goodwill impairment decisions cluster in calendar time, all reported t-statistics and significance stars are based on standard errors two-way clustered by firm and calendar year; t-statistics are presented in parentheses. For *Knightian Uncertainty*, the wild cluster bootstrap p-values (Roodman et al., 2019; MacKinnon et al., 2021) (imposing the null, 9,999 replications, resampling on the calendar-year dimension) are 0.70, 0.90, and 0.40 in Columns (1)–(3) under Rademacher weights; Webb weights yield the same significance classes, and the coefficient is not statistically distinguishable from zero in any specification. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

Table 3: Knightian Uncertainty in Goodwill Impairment Recognition and Measurement

	1st Stage: Goodwill Impairment Indicator			2nd Stage: Goodwill Impairment		
	(1)	(2)	(3)	(4)	(5)	(6)
Knightian Uncertainty	**−0.0578**	**−0.1698****	**−0.1844****	**0.0741*****	**0.0650*****	**0.0583****
	(**−0.92**)	(**−2.58**)	(**−2.53**)	(**3.36**)	(**2.82**)	(**2.21**)
Risk	−0.0648**	−0.0484*	−0.0253	−0.0151	−0.0150	−0.0387**
	(−2.42)	(−1.84)	(−0.79)	(−1.28)	(−1.31)	(−2.21)
Size	0.0044	0.0180***	0.0147**	−0.0039***	−0.0021**	−0.0034
	(1.52)	(5.31)	(2.35)	(−4.00)	(−2.36)	(−0.95)
Leverage	0.0369**	0.0518***	0.0276	0.0030	0.0053	0.0023
	(2.10)	(2.87)	(1.53)	(0.75)	(1.37)	(0.28)
Book-to-Market	0.0673***	0.0751***	0.0764***	0.0011	0.0025*	0.0031
	(4.76)	(5.51)	(6.76)	(0.87)	(2.01)	(1.39)
AltmanZ	0.0034***	0.0037***	0.0045***	0.0026***	0.0025***	0.0021*
	(3.83)	(4.30)	(5.93)	(6.13)	(6.38)	(1.82)
ROA	−0.2177***	−0.2676***	−0.5005***	−0.2094***	−0.2061***	−0.2399***
	(−4.38)	(−5.54)	(−9.85)	(−18.71)	(−20.68)	(−14.87)
Return	−0.0352***	−0.0329***	−0.0217***	−0.0082***	−0.0079***	−0.0036*
	(−5.88)	(−5.42)	(−4.77)	(−3.96)	(−4.67)	(−1.72)
Sales Growth	−0.0482***	−0.0398***	−0.0114**	0.0123***	0.0129***	0.0165***
	(−6.94)	(−6.42)	(−2.13)	(3.65)	(3.90)	(4.46)
Loss	0.1517***	0.1410***	0.1540***	0.0208***	0.0208***	0.0199***
	(10.32)	(9.98)	(11.23)	(10.06)	(9.88)	(9.08)
CAPX	0.0176***	0.0076***	0.0118***	0.0009	0.0000	0.0044**
	(10.55)	(3.99)	(3.29)	(1.61)	(−0.08)	(2.73)
Goodwill Acquisition	0.0197***	0.0108*	−0.0079	0.0023	0.0012	0.0017
	(3.32)	(1.87)	(−1.28)	(1.16)	(0.58)	(1.15)
CFO Volatility	−0.1849***	−0.1720***	0.0763	−0.1052***	−0.1044***	−0.0863**
	(−3.76)	(−3.62)	(1.44)	(−4.37)	(−4.42)	(−2.66)

Table 3: Knightian Uncertainty in Goodwill Impairment Recognition and Measurement (Cont.)

	1st Stage: Goodwill Impairment Indicator			*2nd Stage: Goodwill Impairment*		
	(1)	(2)	(3)	(4)	(5)	(6)
Analyst Forecasts	−0.0100**	−0.0060	0.0120*	0.0003	0.0010	0.0045*
	(−2.14)	(−1.36)	(1.81)	(0.21)	(0.74)	(2.02)
Forecast Dispersion	0.0009	0.0012	0.0080***	−0.0007**	−0.0006**	−0.0006
	(0.96)	(1.35)	(5.57)	(−2.29)	(−2.34)	(−1.06)
Institutional Holding	0.0277**	0.0258**	0.0011	0.0090***	0.0087***	0.0065
	(2.36)	(2.19)	(0.10)	(3.43)	(3.01)	(1.11)
Big 4	0.0143**	0.0025	0.0184**	0.0029	0.0027	0.0002
	(2.73)	(0.49)	(2.65)	(1.23)	(1.13)	(0.04)
Going Concern	0.0025	0.0012	0.0166	−0.0197**	−0.0179**	−0.0176*
	(0.12)	(0.06)	(0.68)	(−2.59)	(−2.37)	(−1.81)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	No	Yes	No	No	Yes	No
Firm FE	No	No	Yes	No	No	Yes
Obs.	47,420	47,420	47,420	5,152	5,152	5,152
Adj. R^2	0.122	0.140	0.237	0.545	0.556	0.686

Notes: This table reports estimates from a two-stage hurdle model examining the relation between Knightian uncertainty and Level 3 fair value estimates, using goodwill impairments as the setting. In the first stage (Columns (1)–(3)), the dependent variable is goodwill impairment recognition (*Goodwill Impairment Indicator*), equal to one if a firm recognizes a goodwill impairment in fiscal year t and zero otherwise. In the second stage (Columns (4)–(6)), the sample is restricted to firm-years in which a firm recognizes a goodwill impairment. The dependent variable is impairment magnitude (*Goodwill Impairment*), measured as the goodwill impairment charge recognized during the fiscal year divided by lagged total assets. The main independent variable of interest is *Knightian Uncertainty*. Control variables include *Risk, Size, Leverage, Book-to-Market, AltmanZ, ROA, Return, Sales Growth, Loss, CAPX, Goodwill Acquisition, CFO Volatility, Analyst Forecasts, Forecast Dispersion, Institutional Holding, Big 4*, and *Going Concern*. Appendix B contains all variable definitions. All specifications include year fixed effects; Columns (2) and (5) additionally include industry fixed effects, and Columns (3) and (6) include firm fixed effects. All reported t-statistics and significance stars are based on standard errors two-way clustered by firm and calendar year; t-statistics are presented in parentheses. For *Knightian Uncertainty*, the wild cluster bootstrap p-values (Roodman et al., 2019; MacKinnon et al., 2021) (imposing the null, 9,999 replications, resampling on the calendar-year dimension) are 0.396, 0.017, 0.032, 0.002, 0.011, and 0.068 in Columns (1)–(6) under Rademacher weights; Webb weights yield the same significance classes. In Column (5) the bootstrap places the coefficient at the 5% rather than the 1% level, and in Column (6) it is significant only at the 10% level and should be read as borderline. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

Table 4: Discretion, Knightian Uncertainty, and Goodwill Impairment

	Dependent Variable: Goodwill Impairment Recognition			
	(1)	(2)	(3)	(4)
Knightian Uncertainty* × *Post* × *Predetermined Goodwill Exposure	**−0.3940****	**−0.4699*****	**−0.4288*****	**−0.4946*****
	(**−2.68**)	(**−3.84**)	(**−4.63**)	(**−5.36**)
Knightian Uncertainty × *Post*	0.0222	0.0457	0.0348	−0.0119
	(0.16)	(0.32)	(0.23)	(−0.05)
Window (fiscal years around adoption)	±1	±2	±3	±4
Lower-order and other interactions	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Decision-Month FE	Yes	Yes	Yes	Yes
Obs.	4,056	7,492	9,852	9,712
Firms	2,028	1,873	1,642	1,214
Decision-Month Clusters	24	48	72	96

Notes: This table reports linear-probability estimates of how the relation between Knightian uncertainty and goodwill impairment recognition changed around ASU 2011-08, which expanded managers' discretion over the qualitative assessment preceding quantitative impairment testing. The sample comprises firm-years with positive beginning goodwill and assets; *Goodwill Impairment Recognition* equals one when a firm recognizes an impairment during the fiscal year. *Post* equals one for fiscal years beginning on or after the standard's effective date, and *Predetermined Goodwill Exposure* is the within-firm-fixed, standardized average ratio of beginning goodwill to beginning assets over the two pre-adoption fiscal years. The reported coefficient is *Knightian Uncertainty* × *Post* × *Predetermined Goodwill Exposure*; *Knightian Uncertainty* is measured over the preceding twelve months. Columns (1)–(4) use balanced ±1 to ±4 fiscal-year windows around adoption, with Column (2) primary. All include firm and decision-month fixed effects, Table 2 controls, and the lower-order interaction terms. This is an associative interrupted-time-series mechanism test with exposure heterogeneity, not a causal difference-in-differences design. Standard errors are two-way clustered by firm and decision month, with t-statistics in parentheses. In Column (2), the triple interaction's two-way and firm-clustered standard errors are 0.1225 and 0.1540. Null-imposed wild-cluster-bootstrap (Roodman et al., 2019; MacKinnon et al., 2021) p-values (9,999 requested replications; common seed 20260720) are 0.0072 under Webb and 0.0063 under Rademacher weights; 18 and 39 infeasible draws were discarded, leaving 9,981 and 9,960 effective replications, respectively, and the two values are not independent draws. A joint test of the exposure-specific pre-adoption coefficients (events −4, −3, and −2; −1 omitted) does not reject differential pre-trends ($p = 0.666$). At the 25th exposure percentile, a one-standard-deviation increase in Knightian uncertainty corresponds to a 2.84-percentage-point higher post-versus-pre change in recognition ($p = 0.004$); at the 90th percentile it corresponds to a 3.50-percentage-point lower change ($p = 0.017$). The corresponding conditional-magnitude interaction, estimated among 489 recognizers in the primary window, is 0.0488 ($p = 0.60$). ***, **, and * denote 1%, 5%, and 10% significance.

Table 5: Knightian Uncertainty and Bond Valuations: Observed-Price Benchmarks

	Dependent Variable: Markdown_Evaluated Price			*Dependent Variable: Markdown_Recent Price*		
	(1)	(2)	(3)	(4)	(5)	(6)
Knightian Uncertainty	**−0.0023****	**−0.0034****	**−0.0030***	**−0.0048****	**−0.0046*****	**−0.0040***
	(−2.25)	**(−2.06)**	**(−1.97)**	**(−2.62)**	**(−2.96)**	**(−1.99)**
Risk	−0.0301***	−0.0259**	−0.0287**	−0.0401	−0.0544*	−0.0707**
	(−4.29)	(−2.64)	(−2.27)	(−1.54)	(−1.81)	(−2.10)
Size	0.0002***	−0.0001	0.0001	0.0002**	−0.0002	−0.0007
	(3.51)	(−0.24)	(0.25)	(2.39)	(−0.31)	(−1.17)
Leverage	−0.0003	−0.0010	−0.0007	0.0005	0.0000	−0.0002
	(−1.39)	(−1.25)	(−0.71)	(0.84)	(0.03)	(−0.12)
AltmanZ	0.0001***	0.0000	0.0000	0.0000	−0.0003**	−0.0003*
	(3.92)	(−0.28)	(0.19)	(−0.10)	(−2.31)	(−1.82)
Return	0.0000	0.0000	0.0001	0.0004	0.0000	0.0007
	(−0.04)	(−0.06)	(0.32)	(0.48)	(0.06)	(1.09)
Portfolio Weight	−0.0001	0.0000	0.0003**	0.0008**	0.0004	0.0006*
	(−0.29)	(−0.20)	(2.64)	(2.03)	(1.18)	(1.71)
Restricted	0.0022***	0.0007	0.0005	0.0046**	0.0017	0.0000
	(4.13)	(1.47)	(1.13)	(2.26)	(1.43)	(−0.02)
Amihud	−0.0059***	−0.0053***	−0.0046***	0.0056	0.0074	0.0087
	(−3.64)	(−3.44)	(−2.68)	(0.71)	(0.93)	(1.12)
Order Imbalance	0.0001	0.0002*	0.0001	0.0012***	0.0014***	0.0015***
	(1.66)	(1.99)	(1.44)	(3.78)	(4.13)	(4.12)
Days since Last Trade	−0.0004***	−0.0004***	−0.0004***	−0.0020**	−0.0021***	−0.0022***
	(−3.73)	(−4.14)	(−3.87)	(−2.61)	(−2.79)	(−2.99)
Days with Trades	0.0005***	0.0005***	0.0004***	0.0007***	0.0006**	0.0004
	(6.20)	(5.52)	(5.27)	(2.77)	(2.35)	(1.60)

Table 5: Knightian Uncertainty and Bond Valuations: Observed-Price Benchmarks (Cont.)

	Dependent Variable: Markdown_Evaluated Price			*Dependent Variable: Markdown_Recent Price*		
	(1)	(2)	(3)	(4)	(5)	(6)
Coupon	−0.0001*	−0.0001		−0.0002***	−0.0002***	
	(−1.86)	(−1.58)		(−2.75)	(−2.99)	
High Yield	−0.0001	0.0002	−0.0003***	−0.0001	0.0002	0.0005
	(−1.08)	(1.67)	(−2.66)	(−0.33)	(0.49)	(0.76)
Bond Age	0.0000	0.0000	0.0001	0.0000*	−0.0001**	0.0007**
	(−0.39)	(−0.21)	(0.88)	(−1.88)	(−2.13)	(2.12)
Maturity	−0.0001***	−0.0001***	−0.0001	−0.0001	−0.0001	−0.0003
	(−19.86)	(−18.61)	(−0.50)	(−1.15)	(−1.13)	(−0.49)
Interest Accruals	−0.0001	−0.0001	0.0000	0.0000	0.0000	0.0000
	(−0.99)	(−1.24)	(−0.82)	(−0.51)	(−0.49)	(−0.34)
Fund-Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	No	No	Yes	No	No
Firm FE	No	Yes	No	No	Yes	No
Bond FE	No	No	Yes	No	No	Yes
Obs.	141,195	141,195	141,195	142,508	142,508	142,508
Adj. R^2	0.294	0.313	0.375	0.428	0.449	0.516

Notes: This table reports OLS estimates examining the relation between issuer-level Knightian uncertainty and mutual funds' reported corporate bond valuations relative to observed market benchmarks. *Markdown_Evaluated Price* equals the log of the fund-reported valuation minus the log of Bloomberg's evaluated price on the reporting date. *Markdown_Recent Price* equals the log of the fund-reported valuation minus the log of the recent-price benchmark, measured as the midpoint of the buy-side and sell-side volume-weighted average prices on the bond's last trading day in the reporting month. The regressions include the issuer-, holding-, liquidity-, and bond-level controls defined in Appendix B. All specifications include fund-quarter fixed effects. Columns (1) and (4) also include industry fixed effects, Columns (2) and (5) include firm fixed effects, and Columns (3) and (6) include bond fixed effects. Standard errors are two-way clustered by firm and year-month, and t-statistics are presented in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

Table 6: Knightian Uncertainty and Bond Valuations: Latent-Price Benchmark

	Dependent Variable: Markdown_Latent Price	
	(1)	(2)
Knightian Uncertainty	**−0.0700****	**−0.0677****
	(**−2.40**)	(**−2.18**)
Risk	0.9345**	0.9320***
	(2.36)	(2.87)
Size	−0.0134	−0.0056
	(−0.52)	(−0.21)
Leverage	−0.0878	−0.0865
	(−1.28)	(−1.34)
AltmanZ	−0.0022	−0.0016
	(−0.59)	(−0.39)
Return	0.0156	0.0146
	(1.44)	(1.35)
Portfolio Weight	0.0325*	0.0333*
	(1.73)	(1.86)
Restricted	−0.0280	−0.0298
	(−1.21)	(−1.45)
Amihud	0.0605	−0.0254
	(1.03)	(−0.57)
Order Imbalance	0.0039	0.0039
	(1.11)	(1.06)
Days since Last Trade	−0.0046	−0.0035
	(−1.64)	(−1.50)
Days with Trades	0.0017	−0.0001
	(0.57)	(−0.02)
Coupon		−0.0061**
		(−2.01)
High Yield		0.0143
		(1.06)
Bond Age		0.0011
		(0.99)
Maturity		0.0011**
		(2.27)
Interest Accruals	−0.0024	−0.0006
	(−0.92)	(−0.30)
Fund-Quarter FE	Yes	Yes
Firm FE	Yes	Yes
Obs.	20,622	20,622
Adj. R^2	0.255	0.273

Table 6: Knightian Uncertainty and Bond Valuations (Cont.)

Notes: This table reports OLS estimates examining the relation between issuer-level Knightian uncertainty and mutual funds' reported corporate bond valuations relative to an estimated latent-price benchmark. *Markdown_Latent Price* equals the log of the fund-reported valuation minus the log of the estimated latent price. The regressions include the issuer-, holding-, liquidity-, and bond-level controls defined in Appendix B. Column (2) additionally includes *Coupon, High Yield, Bond Age*, and *Maturity.* Both specifications include fund-quarter and firm fixed effects. Standard errors are two-way clustered by firm and year-month, and *t*-statistics are presented in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

Table 7: Knightian Uncertainty and Market Reactions to Impairments

Panel A: Market Reaction to Goodwill Impairment Recognition

	Dependent Variable: abs(CAR(−2,+2))		
	(1)	(2)	(3)
Goodwill Impairment Indicator	0.0102***	0.0040**	−0.0132*
	(4.58)	(2.18)	(−1.90)
Goodwill Impairment Indicator × Knightian Uncertainty	**−0.0482****	**−0.0077**	**0.0199**
	(−2.47)	**(−0.46)**	**(0.95)**
Goodwill Impairment Indicator × Risk	0.0197	0.0020	0.0158
	(0.98)	(0.10)	(0.80)
Knightian Uncertainty	−0.1653***	−0.1314***	−0.1170***
	(−10.62)	(−10.04)	(−8.53)
Risk	0.0279***	0.0037	0.0117
	(3.45)	(0.53)	(1.33)
abs(UE)		0.1613***	−0.1763**
		(8.68)	(−2.23)
abs(UE) × Knightian Uncertainty			−2.3666***
			(−3.85)
abs(UE) × Risk			−0.2942**
			(−2.41)
Controls	No	Yes	Yes
Goodwill Impairment Indicator × Controls	No	No	Yes
abs(UE) × Controls	No	No	Yes
Year FE	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes
Obs.	42,813	42,813	42,813
Adj. R^2	0.199	0.209	0.212

Table 7: Knightian Uncertainty and Market Reactions to Impairments (Cont.)

Panel B: Market Reaction to Goodwill Impairment Magnitude

	Dependent Variable: abs(CAR(−2,+2))		
	(1)	(2)	(3)
Goodwill Impairment	0.1717***	0.1143**	0.1974
	(3.77)	(2.34)	(0.88)
***Goodwill Impairment* ×**	**−2.0175****	**−1.6541***	**−2.4644*****
Knightian Uncertainty	**(−2.69)**	**(−2.02)**	**(−2.84)**
Goodwill Impairment × Risk	−0.7991	−0.7140	−0.0545
	(−1.55)	(−1.24)	(−0.12)
Knightian Uncertainty	−0.1031**	−0.0632	−0.0090
	(−2.32)	(−1.50)	(−0.22)
Risk	0.1201**	0.0751	0.0903
	(2.13)	(1.25)	(1.57)
abs(UE)		0.1606*	0.3476
		(1.95)	(1.09)
abs(UE) × Knightian Uncertainty			−0.9858
			(−0.97)
abs(UE) × Risk			−1.1900**
			(−2.31)
Controls	No	Yes	Yes
Goodwill Impairment × Controls	No	No	Yes
abs(UE) × Controls	No	No	Yes
Year FE	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes
Obs.	4,577	4,577	4,577
Adj. R^2	0.265	0.278	0.288

Notes: This table reports OLS estimates examining whether the magnitude of the stock-market response to goodwill impairments varies with Knightian uncertainty. Panel A examines impairment recognition, and Panel B examines impairment magnitude among firm-years with a positive impairment. The dependent variable, *abs(CAR(−2,+2))*, is the absolute value of the firm's compounded return relative to the CRSP value-weighted market return from two trading days before through two trading days after the earnings announcement. The regressions include *Size, Book-to-Market, Leverage*, and *Loss* as controls where indicated. Appendix B defines all variables. All specifications include firm and year fixed effects. Panel A is estimated on 42,813 firm-years (5,427 firms) and Panel B on 4,577 firm-years (2,227 firms); the corrected total shareholder return affects only sample eligibility for these tests and does not enter the estimated equations. All reported t-statistics and significance stars are based on standard errors two-way clustered by firm and calendar year; t-statistics are presented in parentheses. For the focal interaction, the wild cluster bootstrap p-values (Roodman et al., 2019; MacKinnon et al., 2021) (imposing the null, 9,999 replications, resampling on the calendar-year dimension) are 0.026, 0.65, and 0.34 in Panel A and 0.019, 0.072, and 0.016 in Panel B under Rademacher weights; Webb weights yield the same significance classes. In Panel B, Column (2) is borderline at the 10% threshold, and Column (3), which is significant at the 1% level, is significant at the 5% level under the bootstrap. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

Table 8: Knightian Uncertainty, Corporate Bond Markdowns, and Fund Flows

Panel A: Next-Month Fund Flows

	Future Net Flow		*Future Inflow*		*Future Outflow*	
	(1)	(2)	(3)	(4)	(5)	(6)
Aggregated Markdowns	0.1062**	0.1130**	0.1455***	0.1567***	0.0259	0.0290
	(2.49)	(2.58)	(2.76)	(2.91)	(0.53)	(0.58)
Aggregated Markdowns ×	**−1.3499*** **	**−1.4109*** **	**−1.8009*** **	**−1.8603*** **	**−0.2769**	**−0.2764**
Knightian Uncertainty	**(−2.89)**	**(−2.96)**	**(−4.23)**	**(−4.40)**	**(−0.67)**	**(−0.66)**
Aggregated Knightian Uncertainty	3.1532**	3.0803**	4.2234**	4.2462**	0.3449	0.4620
	(2.19)	(2.09)	(2.43)	(2.39)	(0.35)	(0.46)
Aggregated Markdowns × Risk	−2.3525	−2.5666	−4.0998*	−4.1525*	−1.4389	−1.2237
	(−1.21)	(−1.28)	(−1.73)	(−1.77)	(−1.09)	(−0.89)
Aggregated Risk	−18.4128*	−17.6188*	−2.4649	−0.2294	18.3543**	19.7273**
	(−1.97)	(−1.90)	(−0.21)	(−0.02)	(2.45)	(2.58)
Fund Performance	−0.0671***	−0.0635***	−0.1065***	−0.0983***	−0.0351	−0.0309
	(−3.08)	(−2.96)	(−3.50)	(−3.60)	(−1.34)	(−1.24)
Fund Size	−0.0012***	−0.0012***	0.0004	0.0000	0.0015***	0.0012***
	(−3.85)	(−4.02)	(0.83)	(0.10)	(4.15)	(3.54)
Short-Term Leverage	−0.0086	−0.0156**	−0.0394***	−0.0737***	−0.0294***	−0.0543***
	(−1.22)	(−2.32)	(−3.07)	(−5.79)	(−2.80)	(−5.94)
Long-Term Leverage	−0.0393*	−0.0400*	−0.2190***	−0.2072***	−0.1790***	−0.1679***
	(−1.97)	(−1.93)	(−10.63)	(−8.93)	(−14.81)	(−13.01)
Year-Month FE	Yes	Yes	Yes	Yes	Yes	Yes
Fund Family FE	No	Yes	No	Yes	No	Yes
Obs.	16,487	16,487	16,487	16,487	16,487	16,487
Adj. R^2	0.028	0.032	0.028	0.040	0.050	0.062

Table 8: Knightian Uncertainty, Corporate Bond Markdowns, and Fund Flows (CONT.)

Panel B: Next-Quarter Fund Flows

	Future Net Flow		*Future Inflow*		*Future Outflow*	
	(1)	(2)	(3)	(4)	(5)	(6)
Aggregated Markdowns	0.2722**	0.2866**	0.4273***	0.4563***	0.0978	0.1082
	(2.29)	(2.30)	(2.72)	(2.87)	(0.76)	(0.84)
Aggregated Markdowns ×	**−3.0116*** **	**−3.1267*** **	**−5.0419*** **	**−5.1247*** **	**−1.7707**	**−1.7424**
Knightian Uncertainty	(**−2.81**)	(**−2.77**)	(**−3.69**)	(**−3.87**)	(**−1.33**)	(**−1.31**)
Aggregated Knightian Uncertainty	6.4326**	6.1856*	11.0073**	11.0298**	3.4885	3.7632
	(2.06)	(1.98)	(2.33)	(2.32)	(1.16)	(1.25)
Aggregated Markdowns × Risk	−6.3973	−6.4238	−12.8260*	−12.8306*	−4.4050	−4.2433
	(−1.03)	(−1.04)	(−1.81)	(−1.86)	(−0.98)	(−0.97)
Aggregated Risk	−36.0134*	−35.7955*	8.1443	15.3275	41.0709**	46.8868**
	(−1.72)	(−1.70)	(0.32)	(0.60)	(2.34)	(2.60)
Fund Performance	−0.1600**	−0.1472**	−0.3162***	−0.2841***	−0.1403**	−0.1242*
	(−2.35)	(−2.30)	(−2.89)	(−2.94)	(−2.07)	(−1.99)
Fund Size	−0.0059***	−0.0062***	−0.0017	−0.0028**	0.0036***	0.0029***
	(−6.22)	(−6.49)	(−1.20)	(−2.07)	(3.74)	(3.05)
Short-Term Leverage	−0.0341*	−0.0576***	−0.1375***	−0.2489***	−0.0973***	−0.1703***
	(−1.68)	(−3.07)	(−3.48)	(−7.22)	(−3.25)	(−7.55)
Long-Term Leverage	−0.1232**	−0.1195*	−0.7439***	−0.7129***	−0.6159***	−0.5929***
	(−2.01)	(−1.86)	(−11.10)	(−9.65)	(−15.69)	(−14.81)
Year-Month FE	Yes	Yes	Yes	Yes	Yes	Yes
Fund Family FE	No	Yes	No	Yes	No	Yes
Obs.	17,214	17,214	17,214	17,214	17,214	17,214
Adj. R^2	0.041	0.047	0.032	0.049	0.053	0.069

Table 8: Knightian Uncertainty, Corporate Bond Markdowns, and Fund Flows (Cont.)

Notes: This table reports OLS estimates examining how subsequent fund flows vary when a fund's corporate bond markdowns are concentrated in issuers with greater Knightian uncertainty. Panel A examines fund flows during the first month reported in the fund's next available Form N-PORT filing, and Panel B examines fund flows during all three months reported in that filing. The dependent variables are net flows, inflows, and outflows, each divided by the fund's net assets reported in the current filing, where net assets are taken from the most recent amended Form N-PORT filing for each fund-quarter. Net flows equal subscriptions plus reinvestment flows minus redemptions, inflows equal subscriptions plus reinvestment flows, and outflows equal redemptions. In Panel B, the flow measures use the fund's next calendar-quarter filing, are winsorized at the 1st and 99th percentiles of their pooled distributions, and the sample comprises the 17,214 fund-quarters with an adjacent next-quarter filing. *Aggregated Markdowns* × *Knightian Uncertainty* measures the extent to which a fund's corporate bond markdowns are concentrated in issuers with greater Knightian uncertainty. The regressions include the fund-level controls defined in Appendix B. All specifications include year-month fixed effects, and Columns (2), (4), and (6) also include fund-family fixed effects. Standard errors are two-way clustered by fund and year-month, and t-statistics are presented in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

Appendix A: Knightian uncertainty and aversion to Knightian uncertainty

Risk and Knightian uncertainty both concern uncertain future outcomes, but they differ in what is known about the probabilities governing those outcomes. Under risk, uncertainty is represented by a single probability distribution, even though the realized outcome remains unknown. Under Knightian uncertainty, the available information does not identify a unique probability distribution, so multiple distributions remain plausible.

Suppose a firm's future payoff is high ($H = \$100$) or low ($L = \$0$). Under risk, each outcome has a 50% probability. Under Knightian uncertainty, the information may support multiple probability distributions, such as $[70\%, 30\%]$ or $[30\%, 70\%]$.

A decision-maker who is averse to Knightian uncertainty places greater weight on plausible probability distributions that imply unfavorable outcomes and less weight on those that imply favorable outcomes (Gilboa and Schmeidler, 1989; Epstein and Schneider, 2003). The following examples compare how Knightian-uncertainty-neutral and Knightian-uncertainty-averse decision-makers evaluate the same two plausible probability distributions.

A.1. Knightian-uncertainty-neutral decision-maker

A Knightian-uncertainty-neutral decision-maker assigns equal weight to the two plausible probability distributions. The mean and variance of the probability assigned to each outcome, and the resulting measure of Knightian uncertainty ($\mho^2$), are

$$\mathrm{E}[\varphi(H)] = \mathrm{E}[\varphi(L)] = 0.5(0.7) + 0.5(0.3) = 0.5. \tag{A1}$$

$$\mathrm{Var}[\varphi(H)] = \mathrm{Var}[\varphi(L)] = 0.5(0.7 - 0.5)^2 + 0.5(0.3 - 0.5)^2 = 0.04. \tag{A2}$$

$$\mho^2 = 0.5(0.04) + 0.5(0.04) = 0.04. \tag{A3}$$

A.2. Knightian-uncertainty-averse decision-maker

In the expected utility with uncertain probabilities (EUUP) framework, attitudes toward Knightian uncertainty are applied to uncertain probabilities to form certainty-equivalent perceived probabilities (Izhakian, 2017). The perceived probability assigned to the favorable outcome is approximated by

$$\mathrm{E}[\varphi(H)] \left(1 + \frac{\Upsilon''(\cdot)}{\Upsilon'(\cdot)} \operatorname{Var}[\varphi(H)]\right). \tag{A4}$$

Here, Υ is the outlook function applied to probabilities, and

$$\eta = -\frac{\Upsilon''(\cdot)}{\Upsilon'(\cdot)} \tag{A5}$$

denotes the coefficient of absolute Knightian uncertainty aversion.

With constant absolute Knightian uncertainty aversion and $\eta = 4$, the perceived probability of H is approximated by

$$\mathrm{E}[\varphi(H)] \left(1 - \eta \operatorname{Var}[\varphi(H)]\right) = 0.5\,[1 - 4(0.04)] = 0.42. \tag{A6}$$

Holding $\mathrm{E}[\varphi(H)]$ constant, the perceived probability of H decreases as η or $\operatorname{Var}[\varphi(H)]$ increases. Because $\mho^2 = \operatorname{Var}[\varphi(H)]$ in this symmetric binary example, it also decreases with Knightian uncertainty (Izhakian, 2017, 2020).

Appendix B: Variable definitions

Variables	Definition
Knightian Uncertainty	The mean of the monthly firm-level ambiguity estimates provided by Izhakian during the firm's fiscal year, divided by 1,000. The monthly estimates are constructed from five-minute TAQ returns and capture variation in the probabilities assigned to return outcomes. See Izhakian (2017, 2020) for the theoretical foundation.
Risk	The firm's average daily return variance during a month, averaged across the months in its fiscal year. The monthly estimates are provided by Izhakian and constructed from the same five-minute TAQ returns used to estimate Knightian uncertainty.
Goodwill Impairment	The goodwill impairment charge recognized during the fiscal year divided by lagged total assets.
Goodwill Impairment Indicator	Indicator equal to 1 if the firm recognizes a goodwill impairment during the fiscal year and 0 otherwise.
Post	Indicator variable equal to 1 for fiscal years beginning on or after the effective date of ASU 2011-08, and 0 otherwise.
Predetermined Goodwill Exposure	The within-firm-fixed, standardized average ratio of beginning goodwill to beginning assets over the two pre-adoption fiscal years.
Markdown_Evaluated Price	The markdown relative to the Bloomberg evaluated price, calculated as ln(fund-reported valuation) − ln(BVAL).
Markdown_Recent Price	The markdown relative to a recent-price benchmark, calculated as ln(fund-reported valuation) − ln(recent-price benchmark). The benchmark is the midpoint of the buy-side and sell-side volume-weighted average prices on the bond's last trading day in the reporting month.
Markdown_Latent Price	The markdown relative to the estimated latent price, calculated as ln(fund-reported valuation) − ln(latent price). The latent price is estimated separately within each issuer-month using the prices, coupon rates, high-yield status, maturities, and ages of liquid bonds. A bond is classified as liquid if it trades on the report date or its most recent trade occurred no more than five calendar days earlier. We estimate the model only for issuer-months with at least 20 liquid bonds and apply it only within the maturity range represented by those bonds.
Size	The natural logarithm of the firm's total assets in a given quarter or year.

Appendix B: Variable definitions (Cont.)

Variables	Definition
Leverage	The leverage of the firm in a given quarter or year, defined as total liabilities divided by total assets.
Book-to-Market	The firm's book equity, measured as common equity plus deferred taxes (CEQ + TXDB), divided by fiscal-year-end market value of equity (PRCC_F × CSHO).
Book Greater than Market	Indicator variable equal to 1 if the firm's book-to-market ratio exceeds 1.0, and 0 otherwise.
AltmanZ	The modified Altman Z-score, calculated as $1.2(CA/TA) + 1.4(RE/TA) + 3.3(EBIT/TA) + 0.6(MVE/TL) + 0.99(Sales/TA)$. Missing values of current assets and retained earnings are set to zero. The quarterly analyses use the prior fiscal year's annual score.
ROA	The firm's net income divided by average total assets, measured using total assets at the beginning and end of the fiscal period.
Return	The firm's total shareholder return over the fiscal period, constructed from CRSP monthly returns. Using the dated CRSP–Compustat link, we compound the dividend-inclusive, delisting-adjusted monthly total returns over the twelve months (annual analyses) or three months (quarterly analyses) ending in the month of the fiscal-period end.
Sales Growth	The change in the firm's annual sales, calculated as current sales divided by sales in the firm's previous available Compustat observation, minus one.
Loss	Indicator variable equal to 1 if the net income is negative in a given year, and 0 otherwise.
CAPX	The natural logarithm of one plus the firm's capital expenditure during the fiscal year.
Goodwill Acquisition	Indicator variable equal to 1 if the firm records an increase in goodwill due to M&A in a given year, and 0 otherwise.

Appendix B: Variable definitions (CONT.)

Variables	**Definition**
CFO Volatility	For each observation, we first divide operating cash flow by average total assets, measured using total assets from the current and previous available Compustat observations. *CFO Volatility* is the standard deviation of this scaled measure across the current and four previous available observations and requires at least three nonmissing observations.
Analyst Forecasts	The natural logarithm of one plus the number of analyst earnings forecasts for the firm's fiscal year.
Forecast Dispersion	The standard deviation of analysts' most recent annual EPS forecasts for the firm's fiscal year, requiring forecasts from at least three analysts.
Institutional Holding	Institutional ownership at fiscal year-end, calculated as aggregate shares reported by institutional investors in the most recent Form 13F quarter divided by shares outstanding. Holdings are carried forward for no more than two months.
Big 4	Indicator variable equal to 1 if the financial statements are audited by a big four auditor in a given year, and 0 otherwise.
Going Concern	Indicator variable equal to 1 if the auditor issues the going concern opinion in a given year, and 0 otherwise.
Portfolio Weight	The percentage of the fund's net assets invested in the bond on the reporting date.
Restricted	Indicator variable equal to 1 if the holding bond is a restricted security, and 0 otherwise.
Amihud	The mean daily absolute bond return divided by trading volume in millions during the most recent month with trades before the reporting date.
Order Imbalance	Total buy volume minus total sell volume, divided by their sum, during the most recent month with trades before the reporting date.
Days since Last Trade	The natural logarithm of one plus the number of calendar days between the bond's last trade and the quarterly reporting date.

Appendix B: Variable definitions (Cont.)

Variables	**Definition**
Days with Trades	We measure bond trading frequency (*Days with Trades*) using the number of trading days in the bond's most recent month with trades before the reporting date. The measure is the natural logarithm of one plus this count.
Coupon	Coupon rate of a given bond.
High Yield	Indicator variable equal to 1 if the rating of a holding bond is non-investment grade in a given month, and 0 otherwise.
Bond Age	The number of years between the bond's offering date and the beginning of the reporting month.

Appendix B: Variable definitions (Cont.)

Variables	**Definition**
Maturity	The number of years between the beginning of the reporting month and the bond's maturity date.
Interest Accruals	We measure accrued interest (*Interest Accruals*) as the bond's annual coupon rate multiplied by the number of days between its most recent coupon payment and the reporting date, divided by 360.
Aggregated Markdowns	We measure a fund's exposure to bond markdowns (*Aggregated Markdowns*) as the sum, across its stale bond holdings, of each bond's evaluated-price markdown multiplied by the bond's share of fund net assets. We multiply this sum by −1,000 for ease of interpretation.
Aggregated Markdowns × Knightian Uncertainty	We measure the extent to which a fund's bond markdowns are concentrated in issuers with greater Knightian uncertainty (*Aggregated Markdowns × Knightian Uncertainty*) as follows. For each stale bond holding, we multiply the evaluated-price markdown by the issuer's Knightian uncertainty and the bond's share of fund net assets, and then sum these products across the fund. We multiply this sum by −1,000 for ease of interpretation.
Aggregated Knightian Uncertainty	We measure a fund's exposure to issuers' Knightian uncertainty (*Aggregated Knightian Uncertainty*) as the sum, across its stale bond holdings, of each issuer's Knightian uncertainty multiplied by the bond's share of fund net assets.
Aggregated Markdowns × Risk	We measure the extent to which a fund's bond markdowns are concentrated in issuers with greater risk (*Aggregated Markdowns × Risk*) as follows. For each stale bond holding, we multiply the evaluated-price markdown by the issuer's risk and the bond's share of fund net assets, and then sum these products across the fund. We multiply this sum by −1,000 for ease of interpretation.
Aggregated Risk	We calculate portfolio exposure to issuer risk (*Aggregated Risk*) by multiplying the issuer's *Risk* measure for each stale bond holding by that bond's share of fund net assets and summing across the fund.
Fund Performance	We measure fund performance (*Fund Performance*) as the fund's net realized gains plus net unrealized appreciation on nonderivative investments during the quarter, divided by net assets in the previous available fund report.

Appendix B: Variable definitions (CONT.)

Variables	**Definition**
Fund Size	The natural logarithm of fund net assets, taken from Form N-PORT. When a fund files an amended report that corrects a previously reported net-asset value, we use the amendment-resolved value.
Short-Term Leverage	We measure the fund's short-term leverage (*Short-Term Leverage*) as borrowings payable within one year divided by net assets.
Long-Term Leverage	We measure the fund's long-term leverage (*Long-Term Leverage*) as borrowings payable after one year divided by net assets.
Future Net Flow	We measure subsequent net fund flows (*Future Net Flow*) as investor subscriptions plus reinvestment flows minus redemptions, divided by net assets in the current filing. Panel A uses flows during the first month reported in the fund's next available Form N-PORT filing, whereas Panel B sums flows during all three months reported in that filing.
Future Inflow	We measure subsequent fund inflows (*Future Inflow*) as investor subscriptions plus reinvestment flows, divided by net assets in the current filing. Panel A uses flows during the first month reported in the fund's next available Form N-PORT filing, whereas Panel B sums flows during all three months reported in that filing.
Future Outflow	We measure subsequent fund outflows (*Future Outflow*) as redemptions divided by net assets in the current filing. Panel A uses flows during the first month reported in the fund's next available Form N-PORT filing, whereas Panel B sums flows during all three months reported in that filing.

Appendix B: Variable definitions (Cont.)

Variables	Definition
abs(CAR(−2,+2))	We measure the magnitude of the stock-market response (*abs(CAR(−2,+2))*) as the absolute value of the firm's compounded return relative to the CRSP value-weighted market return from two trading days before to two trading days after the earnings announcement. Announcements made at or after 4:00 p.m. are assigned to the next trading day.
abs(UE)	We measure the magnitude of the earnings surprise (*abs(UE)*) as the absolute value of unexpected earnings. Unexpected earnings equals actual EPS minus the most recent preannouncement I/B/E/S median forecast, divided by the split-adjusted stock price at the fiscal-period end. If the fiscal-period end is not a trading day, we use the most recent prior trading day.

Appendix C: Heckman selection model

Table A1: Heckman Selection Model

	First Stage	*Second Stage*
	(1)	(2)
Knightian Uncertainty	**−0.2069**	**0.0702*****
	(0.3211)	**(0.0190)**
Book Greater than Market	0.1213***	
	(0.0371)	
Controls	Yes	Yes
Year FE	Yes	Yes
Industry FE	No	No
Firm FE	No	No
Inverse Mills ratio (λ)		0.0390**
		(0.0189)
Obs.	47,420	5,152

Notes: This table reports estimates from a Heckman two-step selection model examining the relation between Knightian uncertainty and Level 3 fair value estimates in the goodwill impairment setting. The selection equation (first stage) uses a probit model in which the dependent variable is goodwill impairment recognition (*Goodwill Impairment Indicator*), equal to one if a firm recognizes an impairment during the fiscal year and zero otherwise. The outcome equation (second stage) models the magnitude of goodwill impairment among firm-years in which a firm recognizes an impairment. The dependent variable (*Goodwill Impairment*) equals the recognized goodwill impairment charge divided by total assets at the beginning of the fiscal year. The selection equation includes an indicator equal to one when the firm's book-to-market ratio exceeds 1.0 (*Book Greater than Market*) as an exclusion restriction. This condition identifies firms for which book value exceeds market value, providing a market-based signal that the firm may proceed to quantitative impairment testing. The inverse Mills ratio (λ) is reported. The main independent variable of interest is *Knightian Uncertainty.* Control variables include *Risk, Size, Leverage, Book-to-Market, AltmanZ, ROA, Return, Sales Growth, Loss, CAPX, Goodwill Acquisition, CFO Volatility, Analyst Forecasts, Forecast Dispersion, Institutional Holding, Big 4*, and *Going Concern.* Appendix B contains all variable definitions. All specifications include year fixed effects. We obtain standard errors using a firm-clustered bootstrap with 5,000 replications, applied uniformly across all coefficients, and report them in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels. The second-stage estimates use the 5,152 firm-years in which an impairment is recognized.